Sliced Wasserstein distances and their application to regression problems

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Abstract

The sliced Wasserstein distance is a fairly recently developed metric between probability distributions. In this Master's thesis, I investigate how it can be used in conjunction with Gaussian process regression to solve regression problems in which the inputs are represented by probability distributions. After spelling out the theoretical foundations (Wasserstein and sliced Wasserstein distances, positive-definite kernels, and Gaussian process regression), I present a toy example that demonstrates that a workflow incorporating sliced Wasserstein distances can capitalise on distributional differences in the inputs that remain invisible to methods relying solely on marginal distributions and bivariate correlations. I then analyse how global affine transformations of the input distributions affect sliced Wasserstein distances and the kernels derived from them. In particular, I show that different whitening procedures produce the same sliced Wasserstein distances. Asymptotic analyses reveal that extremely anisotropic transformations can lead to degenerate kernel matrices, but that hyperparameter tuning can offset this degeneracy. Simulations and a real-data example corroborate these findings and further suggest that kernels based on Wasserstein distances along the cardinal axes can match or even outperform sliced Wasserstein distances, provided that the predictive signal is approximately axis-aligned.

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Ik draag deze thesis op aan Charlotte Gooskens (1962–2026) en Vincent van Heuven (1949–2026), bij wie ik mijn eerste stappen in de wetenschap zette. Ze hebben mijn leven meer beïnvloed dan ze zelf voor mogelijk hadden kunnen houden.

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Chapter 1

Introduction

This thesis is about methods for quantifying the similarity and dissimilarity between probability distributions and about how these methods can be used to solve prediction problems in statistics and machine learning. While my focus is on regression problems, much of what I will discuss is relevant in classification settings, too.

1.1 Motivation and goal

The most widely used tools for solving classification and regression problems in statistics and machine learning are vector methods. In these methods, the input data are represented as feature vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^d$, and the goal is to learn a function $\widehat{f} : \mathbb{R}^d \to \mathbb{R}$ such that the outputs $y_1, \ldots, y_n \in \mathbb{R}$ are well approximated by $\widehat{f}(\boldsymbol{x}_i), i = 1, \ldots, n$. Several important data types, however, are not naturally expressed as Euclidean vectors. For instance, molecules, social networks, and syntactic structures are more naturally expressed as graphs, whereas a text containing m words may be expressed as an $m \times d$ matrix, the i-th row of which is a vector containing information about the i-th word in the text.

If analysts want to nonetheless apply vector methods when dealing with such objects, they commonly resort to summarising the information contained in these objects numerically and combine these quantities into a feature vector. In the field of graph-based pattern recognition, for instance, vector representations of input graphs may be obtained by computing topological indices (e.g., the sum of the lengths of the shortest paths between all node pairs) or representations based on an eigendecomposition of the adjacency matrices (see Riesen, 2025, Chapter 5 for a concise overview of these techniques). In analyses of text corpora, individual texts could be represented using summaries such as the proportion of nouns that they contain or the mean frequency with which the words contained in them occur in a reference corpus.

While convenient, such summaries inevitably discard some of the information contained in the original objects. A text represented by the mean word frequency of its constituent words loses all information about the distribution of word frequencies within the text; two texts with identical mean word frequencies but entirely different distributional shapes will be treated as indistinguishable. More generally, summarising an $m \times d$ matrix by a fixed-length feature vector requires selecting a particular set of summary statistics, and there is no guarantee that the statistics chosen capture the aspects of the distribution most relevant to the downstream prediction task.

An enticing alternative is to represent the input objects as probability distributions.[1] An example in the field of graph-based pattern recognition is provided by Carpintero Perez et al. (2024, 2025). Without getting into the weeds of graph representations, which we will not need in what follows, these authors encode the input graphs as empirical distributions over multivariate node embeddings. They then express the degree of dissimilarity between these input graphs using a metric on the space of (square-integrable) probability distributions, namely the sliced Wasserstein distance. This distance is subsequently put through a kernel function and fed to a Gaussian process regression model, allowing predictions for test data to be generated. (The terms 'sliced Wasserstein distance', 'kernel function' and 'Gaussian process regression' will be defined precisely in the next chapters.)

Motivated by Carpintero Perez et al.'s approach, the present thesis studies the usefulness of combining sliced Wasserstein distances with Gaussian process regression for regression problems in which the inputs are represented as probability distributions. More specifically, this thesis is built around two questions:

1. Can examples be identified in which the combination of sliced Wasserstein distances and Gaussian process regression substantially outperforms natural alternative approaches even outside of the graph domain covered by Carpintero Perez et al. (2024, 2025)?

2. Rescaling a single dimension can drastically alter pairwise sliced Wasserstein distances, making them scale-sensitive in a way that many vector-based methods are not. How should analysts preprocess distributions to address this sensitivity, and can predictive accuracy be improved by deliberately rescaling informative dimensions?

The first question addresses whether there is a use case for the sliced Wasserstein–Gaussian processes workflow outside of the graph domain. (I was somewhat skeptical at the outset.) The second leads to what are, to my knowledge, my main theoretical contributions as well as to a handful of practical consequences.

1.2 Outline and contributions

Chapter 2 introduces Wasserstein and sliced Wasserstein distances as measures of dissimilarity between probability distributions. It reviews their key theoretical properties and discusses computational aspects. Chapter 3 then presents the framework of Gaussian process regression and shows how Wasserstein and sliced Wasserstein distances can be used in this framework. Since this is achieved by building kernels based on the computed distances, this chapter also introduces the notion of a kernel function. Together, Chapters 2 and 3 lay the methodological foundation for the analyses that follow.

[1] This approach is sometimes referred to as **distribution regression** (e.g., Poczos et al., 2013), though the same term is also used when the outcomes rather than the inputs are distributions (e.g., Klein, 2024).

In Chapter 4, a specially constructed example is used to demonstrate that kernels based on sliced Wasserstein distances can detect distributional differences that remain invisible to conventional vector-based methods as well as to kernels that are based only on marginal distributions. This chapter answers the first question outlined above in the affirmative.

When going beyond such toy examples, though, analysts seeking to use sliced Wasserstein distances to solve real-life regression problems need to deal with a number of practical questions. One such question is whether and how the input distributions should be preprocessed before submitting them to the distance computations. Chapter 5 therefore analyses how global affine transformations, such as rescaling and whitening, affect sliced Wasserstein distances and the kernels derived from them. To my knowledge, mine is the first such treatment. The theoretical results are illustrated by means of simulations. A real-data example further demonstrates how global affine transformations affect the downstream prediction task. Chapter 5 suggests that rescaling informative dimensions can improve predictive accuracy up to a point. It also proves that sliced Wasserstein distances are invariant to certain preprocessing decisions such as the precise whitening technique that is used.

Chapter 6 concludes by summarising the main findings and suggesting avenues for future work that may make for interesting Master's thesis topics.

1.3 Notation and conventions

The space of probability distributions on $\mathbb{R}^d, d \geq 1$, is denoted $\mathcal{P}(\mathbb{R}^d)$. For $p \in [1, \infty)$, $\mathcal{P}_p(\mathbb{R}^d)$ represents the set of probability distributions on $\mathbb{R}^d$ with finite p-th moment. That is,

$$\mathcal{P}_p(\mathbb{R}^d) := \left\{ \mu \in \mathcal{P}(\mathbb{R}^d) : \int_{\mathbb{R}^d} \|x\|_p^p \mathrm{d}\mu(x) < \infty \right\},$$

where

$$\|\boldsymbol{x}\|_p = \|(x_1, \ldots, x_d)^\top\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{\frac{1}{p}}$$

is the p-norm on $\mathbb{R}^d$. The 2-norm on $\mathbb{R}^d$ corresponds to the usual Euclidean norm and may be written as $\|\cdot\|$, dropping the subscript 2.

For probability distributions $\mu \in \mathcal{P}(\mathbb{R}^k), \nu \in \mathcal{P}(\mathbb{R}^\ell), k, \ell \geq 1$, we denote by $\Gamma(\mu, \nu)$ the set of **couplings** of μ and ν. These are the probability distributions on $\mathbb{R}^k \times \mathbb{R}^\ell$ that have μ and ν as their marginals. That is, $\Gamma(\mu, \nu)$ is defined as

$$\left\{ \gamma \in \mathcal{P}(\mathbb{R}^k \times \mathbb{R}^\ell) : \gamma(A \times \mathbb{R}^\ell) = \mu(A), \gamma(\mathbb{R}^k \times B) = \nu(B) \text{ for all } A \in \mathcal{B}(\mathbb{R}^k), B \in \mathcal{B}(\mathbb{R}^\ell) \right\},$$

where $\mathcal{B}(\mathbb{R}^m)$ is the Borel σ-algebra on $\mathbb{R}^m$. Since the product measure $\mu \otimes \nu$ belongs to $\Gamma(\mu, \nu)$, the set $\Gamma(\mu, \nu)$ is not empty. If $\boldsymbol{X} \sim \mu, \boldsymbol{Y} \sim \nu$, then $(\boldsymbol{X}, \boldsymbol{Y}) \sim \mu \otimes \nu$ means that $\boldsymbol{X}$ and $\boldsymbol{Y}$ are (stochastically) independent.

The Dirac measure that places all probability mass on $\boldsymbol{x} \in \mathbb{R}^d$ is denoted $\delta_{\boldsymbol{x}}$.

The unit sphere in $\mathbb{R}^d$ is

$$\mathbb{S}^{d-1} := \{\boldsymbol{\theta} : \|\boldsymbol{\theta}\|_2 = 1\} \subset \mathbb{R}^d.$$

The uniform distribution on $\mathbb{S}^{d-1}$ is denoted as σ.

For a measurable map $f : \mathbb{R}^k \to \mathbb{R}^\ell, k, \ell \geq 1$, we denote as $f_\# : \mathcal{P}(\mathbb{R}^k) \to \mathcal{P}(\mathbb{R}^\ell)$ the **push-forward operator** associated with f. For $\mu \in \mathcal{P}(\mathbb{R}^k)$, $f_\#\mu$ is a probability distribution on $\mathbb{R}^\ell$ characterised by

$$f_\#\mu(A) = \mu(f^{-1}(A))$$

for any measurable set $A \subset \mathbb{R}^\ell$. Here,

$$f^{-1}(A) := \{\boldsymbol{x} \in \mathbb{R}^k : f(x) \in A\}.$$

For $\boldsymbol{u} \in \mathbb{R}^d$, we define the map

$$\begin{aligned} \boldsymbol{u}^\star : \mathbb{R}^d &\to \mathbb{R}, \\ \boldsymbol{x} = (x_1, \ldots, x_d)^\top &\mapsto \langle \boldsymbol{u}, \boldsymbol{x} \rangle = \sum_{i=1}^d u_i x_i. \end{aligned}$$

The associated push-forward operator $(\boldsymbol{u}^\star)_\#$ is also written more compactly as $\boldsymbol{u}^\star_\#$.

For a random variable $X \sim \mu$ on $\mathbb{R}$, its distribution function is the map

$$\mathbb{R} \ni t \mapsto F_\mu(t) := F_X(t) := \mathbb{P}(X \leq t),$$

and its quantile function is defined by means of the generalised inverse of the distribution function as the map

$$(0,1) \ni r \mapsto F_\mu^{-1}(r) := F_X^{-1}(r) := \inf\{t \in \mathbb{R} : F_X(t) \geq r\}.$$

The $\mathbf{1}_n$ symbol represents the vector $(1, \ldots, 1) \in \mathbb{R}^n$ and $\boldsymbol{I}_n$ represents the $n \times n$ identity matrix.

Since the δ symbol is used to denote Dirac measures and I find the $\mathbf{1}$ and $\mathbb{1}$ symbols too similar, I use the Iverson bracket to denote indicator functions. That is, for any proposition P,

$$[P] := \begin{cases} 1, & \text{if } P \text{ is true,} \\ 0, & \text{otherwise.} \end{cases}$$

Thus, $[i = j]$ evaluates to 1 if $i = j$ and to 0 if $i \neq j$.

A non-standard convention that I have found useful is to end numbered statements such as definitions and remarks with a $\diamondsuit$ symbol as opposed to putting the whole statement in italics. So, too, in the first definition in this thesis, which clarifies that I use the terms *distance* and *metric* interchangeably.

Definition 1.1 (Distance)**.** Let $\mathcal{X}$ be a non-empty set. A function $d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is called a **distance** (or a **metric**) on $\mathcal{X}$ if it satisfies all of the following axioms for arbitrary $x, y, z \in \mathcal{X}$:

(D1) $d(x, y) \geq 0$, that is, d is non-negative;

(D2) $d(x, y) = 0$ if and only if $x = y$;

(D3) $d(x, y) = d(y, x)$, that is, d is symmetric;

(D4) $d(x, z) \leq d(x, y) + d(y, z)$, that is, d obeys the triangle inequality. ♢

1.4 Software and code availability

I carried out all computations in R (R Core Team, 2025) and wrote my own functions for computing Wasserstein and sliced Wasserstein distances as well as for fitting Gaussian process regression models. I have put together these functions as an R package, `slicer`, that is available from `https://janhove.github.io/slicer`. The R scripts used for the examples, as well as their inputs and outputs, are available alongside the source files for this thesis from `https://github.com/janhove/mscthesis`.

Chapter 2

Wasserstein and sliced Wasserstein distances

The Wasserstein distance is one of the most important and widely used distances between probability distributions. This chapter reviews some key properties of this distance before turning its focus to its cousin, the sliced Wasserstein distance. Attention is restricted to probability distributions on $\mathbb{R}^d$ for $d \geq 1$.

2.1 Wasserstein distances

2.1.1 Definition and key properties

The mathematical field of **optimal transport** studies the problem of finding a coupling $\gamma \in \Gamma(\mu, \nu)$ between two probability distributions μ and ν defined on metric spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively, that minimises the quantity

$$\int_{\mathcal{X}\times\mathcal{Y}} c(x, y) \mathrm{d}\gamma(x, y),$$

where $c : \mathcal{X} \times \mathcal{Y} \to [0, \infty)$ is a non-negative measurable function (a 'cost function'). Such an optimal coupling is also called an optimal **transport plan**. If, under an optimal transport plan, it holds that $\nu = T_{\#}\mu$ for some measurable map $T : \mathcal{X} \to \mathcal{Y}$, then T is called an optimal **transport map**.

Conceptually, the optimal transport plan represents the most efficient way (in terms of the cost function c) to move the probability mass of μ about so that it corresponds to that of distribution ν. If we consider the special case where μ and ν are probability distributions on $\mathbb{R}^d, d \geq 1$, it is natural to base the cost function on a distance defined on $\mathbb{R}^d$. Of particular interest are cost functions of the form

$$c(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|_p^p,$$

where $\|\cdot\|_p$ denotes the p-norm on $\mathbb{R}^d$ for $p \geq 1$. For our purposes, the focus is less on finding a minimiser $\gamma \in \Gamma(\mu, \nu)$ than it is on computing or otherwise estimating the resulting minimum, which results in the Wasserstein distance between μ and ν.

Definition 2.1 (Wasserstein distance)**.** Let $\mu, \nu \in \mathcal{P}(\mathbb{R}^d), d \geq 1$ and $p \in [1, \infty)$. The **Wasserstein distance** of order p (also called the p-Wasserstein distance) between μ and ν is defined as

$$\mathbf{W}_p(\mu, \nu) := \inf_{\gamma \in \Gamma(\mu,\nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_p^p \mathrm{d}\gamma(x, y) \right)^{\frac{1}{p}}. \qquad \diamondsuit$$

This definition can be recast in the following probabilistic terms: For $\mu, \nu \in \mathcal{P}(\mathbb{R}^d), d \geq 1$, we have

$$\mathbf{W}_p(\mu, \nu) = \inf_{\substack{\boldsymbol{X} \sim \mu, \\ \boldsymbol{Y} \sim \nu}} \left(\mathbb{E}(\|\boldsymbol{X} - \boldsymbol{Y}\|_p^p) \right)^{\frac{1}{p}} \tag{2.1}$$

for all $p \geq 1$. The infimum in Definition 2.1 is a minimum in $[0, \infty)$ provided that both μ and ν have finite p-th moments (Villani, 2009, Theorem 4.1 and Chapter 6).

I will often abuse notation and write $\mathbf{W}_p(\boldsymbol{X}, \boldsymbol{Y})$ in lieu of $\mathbf{W}_p(\mu, \nu)$ if $\boldsymbol{X} \sim \mu$ and $\boldsymbol{Y} \sim \nu$. Further, if $\mu = m^{-1} \sum_{i=1}^m \delta_{\boldsymbol{x}_i}$ and $\nu = n^{-1} \sum_{j=1}^n \delta_{\boldsymbol{y}_j}$ are empirical distributions with $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m$ and $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ vectors in $\mathbb{R}^d$, I will occasionally write $\mathbf{W}_p(\boldsymbol{M}_1, \boldsymbol{M}_2)$ instead of $\mathbf{W}_p(\mu, \nu)$, where

$$\boldsymbol{M}_1 = \begin{pmatrix} \boldsymbol{x}_1^\top \\ \vdots \\ \boldsymbol{x}_m^\top \end{pmatrix} \quad \text{and} \quad \boldsymbol{M}_2 = \begin{pmatrix} \boldsymbol{y}_1^\top \\ \vdots \\ \boldsymbol{y}_n^\top \end{pmatrix}.$$

The Wasserstein distance, named for Leonid Vaseršteĭn in what seems to be a case of Stigler's law of eponymy, goes by a few different names, among which are the Mallows distance, the Kantorovich–Rubinstein metric, and the earth mover's distance (for $p = 1$).

A few properties of the Wasserstein distance that I will use in the chapters to follow are presented in the next lemma.

Lemma 2.2 (Basic properties of $\mathbf{W}$)**.** For $p \geq 1$ and $d \geq 1$, let $\boldsymbol{X} \sim \mu, \boldsymbol{Y} \sim \nu$ be random vectors, with $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$. Further let $\boldsymbol{v} \in \mathbb{R}^d$ and $c \in \mathbb{R}$. Then

1. $\mathbf{W}_p(\boldsymbol{X} + \boldsymbol{v}, \boldsymbol{Y} + \boldsymbol{v}) = \mathbf{W}_p(\boldsymbol{X}, \boldsymbol{Y})$, that is, Wasserstein distances are translation-invariant.

2. $\mathbf{W}_p(c\boldsymbol{X}, c\boldsymbol{Y}) = |c| \mathbf{W}_p(\boldsymbol{X}, \boldsymbol{Y})$, that is, Wasserstein distances are homogeneous.

For the special case $p = 2$, furthermore, the Wasserstein distance can be bounded from above:

3. $\mathbf{W}_2^2(\boldsymbol{X}, \boldsymbol{Y}) \leq 2\mathbb{E}(\|\boldsymbol{X}\|^2) + 2\mathbb{E}(\|\boldsymbol{Y}\|^2)$. $\diamondsuit$

Proof. From Equation 2.1, we obtain

$$\begin{aligned}
\mathbf{W}_p(\boldsymbol{X}+\boldsymbol{v},\boldsymbol{Y}+\boldsymbol{v}) &= \inf_{\substack{\widetilde{\boldsymbol{X}}\sim\mathcal{L}(\boldsymbol{X}),\\ \widetilde{\boldsymbol{Y}}\sim\mathcal{L}(\boldsymbol{Y})}} \left(\mathbb{E}(\|\widetilde{\boldsymbol{X}}+\boldsymbol{v}-(\widetilde{\boldsymbol{Y}}+\boldsymbol{v})\|_p^p)\right)^{\frac{1}{p}}\\
&= \inf_{\substack{\widetilde{\boldsymbol{X}}\sim\mathcal{L}(\boldsymbol{X}),\\ \widetilde{\boldsymbol{Y}}\sim\mathcal{L}(\boldsymbol{Y})}} \left(\mathbb{E}(\|\widetilde{\boldsymbol{X}}-\widetilde{\boldsymbol{Y}}\|_p^p)\right)^{\frac{1}{p}}\\
&= \mathbf{W}_p(\boldsymbol{X},\boldsymbol{Y}).
\end{aligned}$$

Similarly,

$$\mathbf{W}_p(c\boldsymbol{X},c\boldsymbol{Y}) = \inf_{\substack{\widetilde{\boldsymbol{X}}\sim\mathcal{L}(\boldsymbol{X}),\\ \widetilde{\boldsymbol{Y}}\sim\mathcal{L}(\boldsymbol{Y})}} \left(\mathbb{E}(\|c\widetilde{\boldsymbol{X}}-c\widetilde{\boldsymbol{Y}}\|_p^p)\right)^{\frac{1}{p}} = |c| \inf_{\substack{\widetilde{\boldsymbol{X}}\sim\mathcal{L}(\boldsymbol{X}),\\ \widetilde{\boldsymbol{Y}}\sim\mathcal{L}(\boldsymbol{Y})}} \left(\mathbb{E}(\|\widetilde{\boldsymbol{X}}-\widetilde{\boldsymbol{Y}}\|_p^p)\right)^{\frac{1}{p}} = |c|\mathbf{W}_p(\boldsymbol{X},\boldsymbol{Y}).$$

Finally, for independent $\widetilde{\boldsymbol{X}}\sim\mathcal{L}(\boldsymbol{X})$ and $\widetilde{\boldsymbol{Y}}\sim\mathcal{L}(\boldsymbol{Y})$, it holds that

$$\mathbf{W}_2^2(\boldsymbol{X},\boldsymbol{Y}) \le \mathbb{E}\left(\|\widetilde{\boldsymbol{X}}-\widetilde{\boldsymbol{Y}}\|^2\right) = \mathbb{E}\left(\|\widetilde{\boldsymbol{X}}\|^2\right) + \mathbb{E}\left(\|\widetilde{\boldsymbol{Y}}\|^2\right) - 2\mathbb{E}(\langle\widetilde{\boldsymbol{X}},\widetilde{\boldsymbol{Y}}\rangle).$$

We find that

$$\begin{aligned}
-2\mathbb{E}(\langle\widetilde{\boldsymbol{X}},\widetilde{\boldsymbol{Y}}\rangle) &= -2\langle\mathbb{E}(\widetilde{\boldsymbol{X}}),\mathbb{E}(\widetilde{\boldsymbol{Y}})\rangle && \text{[independence]}\\
&\le 2|\langle\mathbb{E}(\widetilde{\boldsymbol{X}}),\mathbb{E}(\widetilde{\boldsymbol{Y}})\rangle|\\
&\le 2\|\mathbb{E}(\widetilde{\boldsymbol{X}})\|\|\mathbb{E}(\widetilde{\boldsymbol{Y}})\| && \text{[Cauchy–Schwarz]}\\
&\le \|\mathbb{E}(\widetilde{\boldsymbol{X}})\|^2 + \|\mathbb{E}(\widetilde{\boldsymbol{Y}})\|^2 && [a^2+b^2\ge 2ab]\\
&\le \mathbb{E}(\|\widetilde{\boldsymbol{X}}\|^2) + \mathbb{E}(\|\widetilde{\boldsymbol{Y}}\|^2). && \text{[Jensen]}
\end{aligned}$$

Since $\boldsymbol{X} \overset{d}{=} \widetilde{\boldsymbol{X}}, \boldsymbol{Y} \overset{d}{=} \widetilde{\boldsymbol{Y}}$, it follows that $\mathbf{W}_2^2(\boldsymbol{X},\boldsymbol{Y}) \le 2\mathbb{E}\left(\|\boldsymbol{X}\|^2\right) + 2\mathbb{E}\left(\|\boldsymbol{Y}\|^2\right)$. □

The p-Wasserstein distance is a distance on $\mathcal{P}_p(\mathbb{R}^d), d \ge 1, p \ge 1$, in the sense of Definition 1.1 on page 4 (Villani, 2009, Chapter 6). From a theoretical point of view, one of its attractive properties is that it metrises weak convergence on $\mathcal{P}_p(\mathbb{R}^d)$: $\mathbf{W}_p(\mu_k,\mu) \to 0, k\to\infty$, if and only if μ_k converges weakly to μ (Villani, 2009, Theorem 6.9). Another appealing property, which I will use in a later chapter, is provided in the next lemma.

Lemma 2.3 (Continuity of **W**)**.** The p-Wasserstein distance is continuous on $\mathcal{P}_p(\mathbb{R}^d)$ in the following sense: if $(\mu_k)_{k=1}^\infty, (\nu_k)_{k=1}^\infty$ converge weakly to μ and ν in $\mathcal{P}_p(\mathbb{R}^d)$, respectively, $k\to\infty$, then $\mathbf{W}_p(\mu_k,\nu_k) \to \mathbf{W}_p(\mu,\nu), k\to\infty$. ♢

Proof. Due to the triangle inequality,

$$\begin{aligned}
\mathbf{W}_p(\mu_k,\nu_k) - \mathbf{W}_p(\mu,\nu) &\le \mathbf{W}_p(\mu_k,\mu) + \mathbf{W}_p(\mu,\nu) + \mathbf{W}_p(\nu,\nu_k) - \mathbf{W}_p(\mu,\nu)\\
&= \mathbf{W}_p(\mu_k,\mu) + \mathbf{W}_p(\nu,\nu_k),
\end{aligned}$$

which converges to 0 as $k \to \infty$ since $\mathbf{W}_p$ metrises weak convergence on $\mathcal{P}_p(\mathbb{R}^d)$. Similarly,

$$\begin{aligned}\mathbf{W}_p(\mu, \nu) - \mathbf{W}_p(\mu_k, \nu_k) &\leq \mathbf{W}_p(\mu, \mu_k) + \mathbf{W}_p(\mu_k, \nu_k) + \mathbf{W}_p(\nu_k, \nu) - \mathbf{W}_p(\mu_k, \nu_k) \\ &= \mathbf{W}_p(\mu, \mu_k) + \mathbf{W}_p(\nu_k, \nu),\end{aligned}$$

which also converges to 0 as $k \to \infty$. Hence,

$$|\mathbf{W}_p(\mu_k, \nu_k) - \mathbf{W}_p(\mu, \nu)| \to 0,$$

as $k \to \infty$. □

Since the p-Wasserstein distance is a distance on $\mathcal{P}_p(\mathbb{R}^d)$, we can use it to compute pairwise distances between objects that can be encoded as distributions in $\mathcal{P}_p(\mathbb{R}^d)$. Distance-based procedures such as k-nearest neighbours can subsequently be trained on these distances in order to solve classification and regression problems. However, an important class of predictive models, which includes support vector machines and Gaussian process regression, does not work directly with distances between objects $x_1, \ldots, x_n$. Instead, the pairwise similarity between these objects is expressed by means of a kernel. I leave the discussion of what kernels are and how they can be constructed and used in predictive modelling for Chapter 3.

2.1.2 Computation

In the general case, computing the exact p-Wasserstein distance between two distributions supported on at most n points has time complexity $\mathcal{O}(n^3 \log n)$, whereas an approximate solution can be obtained in $\mathcal{O}(n^2/\epsilon)$ for a precision level $\epsilon > 0$ (Nguyen, 2025, Remark 2.3). However, there are two special cases in which Wasserstein distances can be computed considerably more cheaply.

The first one concerns Wasserstein distances of order $p = 2$ between Gaussian distributions.

Proposition 2.4. Let $\mu_1 = \mathcal{N}_d(\boldsymbol{m}_1, \boldsymbol{\Sigma}_1), \mu_2 = \mathcal{N}_d(\boldsymbol{m}_2, \boldsymbol{\Sigma}_2)$ be Gaussian distributions on $\mathbb{R}^d$ with symmetric positive-definite covariance matrices. Then

$$\mathbf{W}_2^2(\mu_1, \mu_2) = \|\boldsymbol{m}_1 - \boldsymbol{m}_2\|^2 + \operatorname{Tr}\left(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2 - 2(\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{\Sigma}_2\boldsymbol{\Sigma}_1^{1/2})^{1/2}\right). \qquad \diamond$$

Proof. See Dowson & Landau (1982). These authors write their result as

$$\mathbf{W}_2^2(\mu_1, \mu_2) = \|\boldsymbol{m}_1 - \boldsymbol{m}_2\|^2 + \operatorname{Tr}\left(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2 - 2(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)^{1/2}\right).$$

However, since

$$\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma}_1^{1/2}(\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{\Sigma}_2\boldsymbol{\Sigma}_1^{1/2})\boldsymbol{\Sigma}_1^{-1/2},$$

the matrices $\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2$ and $\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{\Sigma}_2\boldsymbol{\Sigma}_1^{1/2}$ are similar. In particular, they have the same eigenvalues, and so do their matrix square roots. Consequently, they have the same trace. Later authors seem to prefer the longer version, perhaps because this makes it easier to appreciate that the distance is symmetric. □

Data: A discrete probability distribution $\mu = \sum_{i=1}^{m} \alpha_i \delta_{x_i}$ on $\mathbb{R}$
A discrete probability distribution $\nu = \sum_{j=1}^{n} \beta_j \delta_{y_j}$ on $\mathbb{R}$
An order $p \geq 1$
Result: $W = \mathbf{W}_p(\mu, \nu)$
1 `NorthWestCorner`(μ, ν, p):
2   Let $(\alpha_{(i)}, x_{(i)})_{i=1}^{m}$ be $(\alpha_i, x_i)_{i=1}^{m}$ sorted in ascending order of $\boldsymbol{x}$.
3   Let $(\beta_{(j)}, y_{(j)})_{j=1}^{n}$ be $(\beta_j, y_j)_{j=1}^{n}$ sorted in ascending order of $\boldsymbol{y}$.
4   $W \leftarrow 0$
5   $i \leftarrow 1$
6   $j \leftarrow 1$
7   $a \leftarrow \alpha_{(1)}$
8   $b \leftarrow \beta_{(1)}$
9   **while** $i \leq m, j \leq n$ **do**
10     $t \leftarrow \min(a, b)$
11     $W \leftarrow W + t|x_{(i)} - y_{(j)}|^p$
12     $a \leftarrow a - t$
13     $b \leftarrow b - t$
14   **if** $a = 0$ **then**
15       $i \leftarrow i + 1$
16   **if** $i \leq m$ **then**
17         $a \leftarrow \alpha_{(i)}$
18   **if** $b = 0$ **then**
19       $j \leftarrow j + 1$
20   **if** $j \leq n$ **then**
21         $b \leftarrow \beta_{(j)}$
22   $W \leftarrow W^{1/p}$
23   **return** W

Algorithm 2.1: The northwest corner algorithm.

The second special case, which is key to this thesis, occurs when $d = 1$.

Proposition 2.5 (Univariate distributions)**.** Let F_{μ}^{-1} and F_{ν}^{-1} be the quantile functions of $\mu, \nu \in \mathcal{P}_p(\mathbb{R})$, respectively. Then

$$\mathbf{W}_p^p(\mu, \nu) = \int_0^1 |F_{\mu}^{-1}(t) - F_{\nu}^{-1}(t)|^p \mathrm{d}t. \qquad \diamondsuit$$

Proof. Nadjahi (2021, p. 21) cites Theorem 3.1.2(a) in Rachev & Rüschendorf (1998) as proof for this fact. $\square$

If the quantile functions of μ and ν are known, the integral in Proposition 2.5 can be evaluated numerically. If μ and ν are both discrete distributions that are supported on m and n points, respectively (e.g., if μ, ν are empirical distributions), an exact value can be computed using the **northwest corner algorithm** (Algorithm 2.1). The *while*-loop executes at most $m + n - 1$ times, so the algorithm's asymptotic time complexity is dominated by the sorting operations in lines 2–3. This means that the algorithm runs in $\mathcal{O}(N \log N)$, where $N = \max(m, n)$.

If

$$\mu = \frac{1}{n}\sum_{i=1}^{n}\delta_{x_i} \quad \text{and} \quad \nu = \frac{1}{n}\sum_{i=1}^{n}\delta_{y_i},$$

with $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$, computing the Wasserstein distance between μ and ν reduces to a straightforward sorting problem with time complexity $\mathcal{O}(n \log n)$:

$$\mathbf{W}_p^p(\mu, \nu) = \frac{1}{n}\sum_{i=1}^{n}|x_{(i)} - y_{(i)}|^p,$$

where $u_{(i)}$ is the i-th order statistic of $\boldsymbol{u}$.

2.2 Sliced Wasserstein distances

Unlike in the general case, Wasserstein distances between univariate distributions are easy to compute, particularly between discrete distributions. This led Rabin et al. (2012) to define a distance between multidimensional probability distributions in terms of the Wasserstein distances between one-dimensional projections of these distributions. This approach, termed sliced Wasserstein distances, has gained considerable traction since then. A recent book-length treatment of sliced Wasserstein distances and further developments is provided by Nguyen (2025).

2.2.1 Definition and properties

To recall, we define the unit sphere of $\mathbb{R}^d$ in terms of the Euclidean norm, that is,

$$\mathbb{S}^{d-1} := \{\boldsymbol{\theta} \in \mathbb{R}^d : \|\boldsymbol{\theta}\| = 1\}.$$

The uniform distribution on $\mathbb{S}^{d-1}$ is denoted as σ. The sliced Wasserstein distance is now formally defined as follows.

Definition 2.6 (Sliced Wasserstein distance)**.** For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d), p \geq 1, d \geq 1$, the **sliced Wasserstein distance** of order p between μ and ν is defined as

$$\mathbf{SW}_p(\mu, \nu) := \left(\int \mathbf{W}_p^p(\theta^\star_\# \mu, \theta^\star_\# \nu)\mathrm{d}\sigma(\theta)\right)^{\frac{1}{p}}. \qquad \diamondsuit$$

In probabilistic terms, the sliced Wasserstein distance can be written as

$$\mathbf{SW}_p(\mu, \nu) = \left(\mathbb{E}_{\boldsymbol{\theta}\sim\sigma}\left(\mathbf{W}_p^p(\boldsymbol{\theta}^\star_\# \mu, \boldsymbol{\theta}^\star_\# \nu)\right)\right)^{\frac{1}{p}}.$$

For $\boldsymbol{X} \sim \mu, \boldsymbol{Y} \sim \nu$, it is often convenient to write

$$\mathbf{SW}_p(\mu, \nu) =: \mathbf{SW}_p(\boldsymbol{X}, \boldsymbol{Y}) = \left(\int \mathbf{W}_p^p(\langle \boldsymbol{X}, \theta\rangle, \langle \boldsymbol{Y}, \theta\rangle)\mathrm{d}\sigma(\theta)\right)^{\frac{1}{p}}.$$

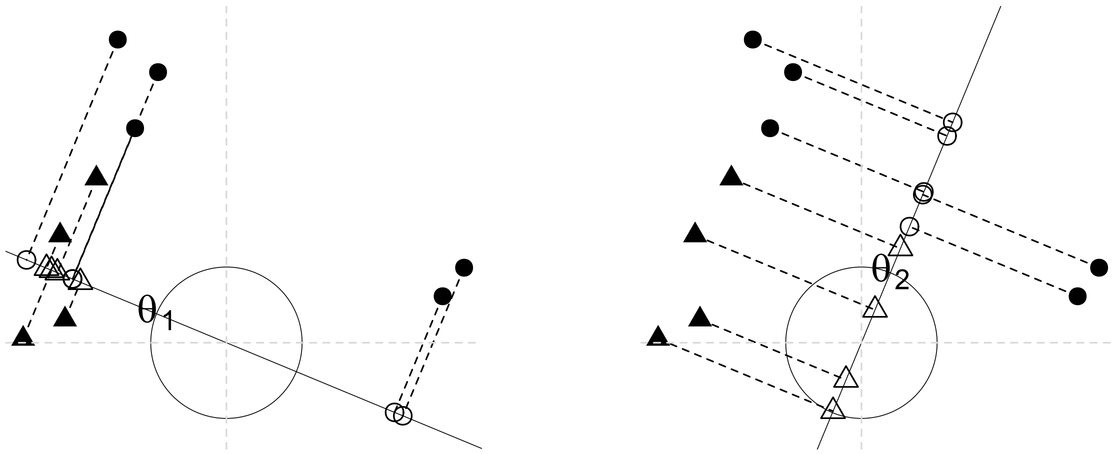

Figure 2.1: The circles and triangles are projected orthogonally onto the span of the unit vectors $\boldsymbol{\theta}_1$ (left) and $\boldsymbol{\theta}_2$ (right). In each case, the p-Wasserstein distance between the projected distributions can easily be computed. The sliced Wasserstein distance of order p is defined in terms of the expectation of the p-Wasserstein distance between such projected distributions for a random vector $\boldsymbol{\theta}$ sampled uniformly from the unit sphere.

Similarly, if μ, ν are empirical distributions represented by matrices $\boldsymbol{M}_1, \boldsymbol{M}_2$, I will occasionally write

$$\mathbf{SW}_p(\mu,\nu) =: \mathbf{SW}_p(\boldsymbol{M}_1, \boldsymbol{M}_2) = \left(\int \mathbf{W}_p^p(\boldsymbol{M}_1\theta, \boldsymbol{M}_2\theta)\mathrm{d}\sigma(\theta)\right)^{\frac{1}{p}}.$$

Figure 2.1 illustrates the intuition behind the definition of the sliced Wasserstein distance.

Like the p-Wasserstein distance, the sliced Wasserstein distance of order p is a distance on $\mathcal{P}_p(\mathbb{R}^d)$ (Bonnotte, 2013, Proposition 5.1.2; see also Nadjahi, 2021, Proposition 7.1, for a more general statement). Further theoretical properties such as the relation between weak convergence and the convergence under $\mathbf{SW}_p$ are discussed by Nadjahi (2021, Chapter 7), but these findings are not relevant for the remainder of this thesis.

2.2.2 Computation and estimation

In practice, sliced Wasserstein distances need to be estimated using Monte Carlo simulation. The use case that is relevant for this thesis is when $\mu = m^{-1}\sum_{i=1}^m \delta_{\boldsymbol{x}_i}$ and $\nu = n^{-1}\sum_{j=1}^n \delta_{\boldsymbol{y}_j}$ are discrete (empirical) distributions on $\mathbb{R}^d$. Representing these distributions in matrix form as

$$\boldsymbol{M}_1 = \begin{pmatrix} \boldsymbol{x}_1^\top \\ \vdots \\ \boldsymbol{x}_m^\top \end{pmatrix} \quad \text{and} \quad \boldsymbol{M}_2 = \begin{pmatrix} \boldsymbol{y}_1^\top \\ \vdots \\ \boldsymbol{y}_n^\top \end{pmatrix},$$

the sliced Wasserstein distance of order p between μ and ν is estimated as

$$\widehat{\mathbf{SW}}_p^p(\mu,\nu;\boldsymbol{\theta}_1,\ldots,\boldsymbol{\theta}_L) := \frac{1}{L}\sum_{\ell=1}^{L} \mathbf{W}_p^p(\boldsymbol{M}_1\boldsymbol{\theta}_\ell, \boldsymbol{M}_2\boldsymbol{\theta}_\ell),$$

with $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L \overset{\text{i.i.d.}}{\sim} \sigma$, $L \geq 1$. Nguyen (2025, Remark 2.17) bounds the Monte Carlo estimation error as

$$\mathbb{E}\left|\widehat{\mathbf{SW}}_p^p(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L) - \mathbf{SW}_p^p(\mu, \nu)\right| \leq \frac{1}{\sqrt{L}} \mathrm{Var}_{\boldsymbol{\theta} \sim \sigma}\left(\mathbf{W}_p^p(\boldsymbol{M}_1 \boldsymbol{\theta}, \boldsymbol{M}_2 \boldsymbol{\theta})\right)^{\frac{1}{2}}.$$

When estimating pairwise sliced Wasserstein distances between N empirical distributions each supported on n points in $\mathbb{R}^d$, the analyst needs to sample L independent vectors $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L$ from σ, compute LN products involving N $n \times d$ matrices and L d-dimensional vectors (time complexity in $\mathcal{O}(LNnd)$), and then sort the resulting LN n-dimensional vectors (time complexity in $\mathcal{O}(LNn \log n)$). On sorted inputs, the northwest corner algorithm runs in linear time, so the final step involves $\mathcal{O}(N^2)$ computations that each have time complexity $\mathcal{O}(Ln)$. Putting everything together, the $N(N-1)/2$ sliced Wasserstein distances can be estimated in

$$\mathcal{O}(LNnd + LNn \log n + N^2 Ln) = \mathcal{O}(LNn(d + \log n + N)).$$

In the next chapter, we will see how Wasserstein and sliced Wasserstein distances can be used in a kernel-based regression framework.

Chapter 3

Kernels and kernel-based regression

This chapter develops the regression framework that I will use in the remainder of the thesis: Gaussian process regression. This method assumes that we have at our disposal a way to express the similarity between the input objects. More precisely, it assumes that this similarity is expressed using a positive-definite kernel.

Definition 3.1 (Positive-definite kernel)**.** Let $\mathcal{X}$ be a non-empty set. A function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is called a **positive-definite kernel** on $\mathcal{X}$ if

(K1) k is symmetric, i.e., if $k(x, x') = k(x', x)$ for all $x, x' \in \mathcal{X}$, and

(K2) for all $x_1, \ldots, x_n \in \mathcal{X}, c_1, \ldots, c_n \in \mathbb{R}, n \geq 1$, it holds that

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j k(x_i, x_j) \geq 0.$$

These conditions are equivalent to requiring that, for any $n \geq 1$ and any choice of $x_1, \ldots, x_n \in \mathcal{X}$, the matrix

$$\boldsymbol{K} = (k(x_i, x_j))_{1 \leq i,j \leq n}$$

is symmetric positive semi-definite. ♢

Given two positive-definite kernels, further ones may be constructed in various ways. The two ways we will use are licensed by the following lemma.

Lemma 3.2. Let $k_1, k_2 : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be two positive-definite kernels on $\mathcal{X}$. Then both their sum $k_1 + k_2$ and their pointwise product $k_1 k_2$ are positive-definite kernels, too. ♢

Proof. The symmetry of $k_1 + k_2$ and $k_1 k_2$ follows immediately from the symmetry of k_1 and k_2.

As for the positive-definiteness of $k_1 + k_2$, consider arbitrary $x_1, \ldots, x_n \in \mathcal{X}, c_1, \ldots, c_n \in \mathbb{R}, n \geq 1$. By the positive-definiteness of k_1 and k_2, it holds that

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j (k_1(x_i, x_j) + k_2(x_i, x_j)) = \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j k_1(x_i, x_j) + \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j k_2(x_i, x_j) \geq 0 + 0,$$

so $k_1 + k_2$ is a positive-definite kernel.

We now consider the pointwise product. For $n \geq 1$, write

$$A = (k_1(x_i, x_j))_{1 \leq i,j \leq n} \quad \text{and} \quad B = (k_2(x_i, x_j))_{1 \leq i,j \leq n}.$$

Both A and B are real symmetric positive semi-definite matrices, so we may write them in terms of an eigenvalue decomposition:

$$A = \sum_{k=1}^{n} a_k u_k u_k^\top \quad \text{and} \quad B = \sum_{\ell=1}^{n} b_\ell v_\ell v_\ell^\top,$$

with eigenvalues $a_k, b_\ell \geq 0, 1 \leq k, \ell \leq n$. Denoting the pointwise (Hadamard) product between A and B as

$$A \odot B := (A_{ij} B_{ij})_{1 \leq i,j \leq n},$$

we obtain, for arbitrary $c \in \mathbb{R}^n$,

$$\begin{aligned}
c^\top (A \odot B) c &= \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j (A \odot B)_{ij} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \left(\sum_{k=1}^{n} a_k u_{ki} u_{kj} \right) \left(\sum_{\ell=1}^{n} b_\ell v_{\ell i} v_{\ell j} \right) \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{k=1}^{n} \sum_{\ell=1}^{n} a_k b_\ell c_i u_{ki} v_{\ell i} c_j u_{kj} v_{\ell j} \\
&= \sum_{k=1}^{n} \sum_{\ell=1}^{n} a_k b_\ell \left(\sum_{i=1}^{n} c_i u_{ki} v_{\ell i} \right) \left(\sum_{j=1}^{n} c_j u_{kj} v_{\ell j} \right) \\
&= \sum_{k=1}^{n} \sum_{\ell=1}^{n} \underbrace{a_k b_\ell}_{\geq 0} \left(\sum_{i=1}^{n} c_i u_{ki} v_{\ell i} \right)^2 \\
&\geq 0.
\end{aligned}$$

Consequently, $A \odot B$ is symmetric positive semi-definite, which means that the pointwise product $k_1 k_2$ is a positive-definite kernel. □

I postpone the discussion of how we can construct such positive-definite kernels by means of Wasserstein and sliced Wasserstein distances to Section 3.2. First, in Section 3.1, I discuss the basics of Gaussian process regression while taking for granted that a suitable positive-definite kernel is available. The chapter closes with a discussion of how the hyperparameters in the kernels and in the Gaussian process regression proper can be optimised via likelihood maximisation.

3.1 Gaussian process regression

An introduction into the theory of Gaussian process regression is provided by Rasmussen & Williams (2006). Here, I only cover the nuts and bolts insofar as they are relevant for the remainder of this thesis. I do not rederive the formulae below myself; I merely reproduce them from Rasmussen & Williams (2006) with slight changes in notation in order to be more consistent with the notation used in this thesis.

Let $\boldsymbol{x} = (x_i)_{i=1}^{N_1} \subset \mathcal{X}$ be input data for which we observe outcomes $y_1, \ldots, y_{N_1} \in \mathbb{R}$. We assume that

$$y_i = f(x_i) + \varepsilon_i, i = 1, \ldots, N_1,$$

with fixed inputs $\boldsymbol{x}$ and i.i.d. random errors $\varepsilon_1, \ldots, \varepsilon_{N_1} \sim \mathcal{N}(0, \eta^2), \eta^2 \geq 0$. The process $f : \mathcal{X} \to \mathbb{R}$ is modelled as a **Gaussian process**, that is, as an indexed collection of random variables $(f(x))_{x \in \mathcal{X}}$ such that $(f(x_1), \ldots, f(x_n))$ has an n-variate normal distribution for any $n \geq 1$. Treating f as stochastic allows us to do inference on its posterior distribution; for present purposes, however, only point predictions are of interest, which are obtained as the posterior mean. I only consider Gaussian processes with a constant mean $m \in \mathbb{R}$, which are fully specified by a covariance function represented by a positive-definite kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}, (x, x') \mapsto \mathrm{Cov}(f(x), f(x'))$.

The data points $(x_i, y_i)_{i=1}^{N_1}$ comprise the training set. We are given a further N_2 elements of $\mathcal{X}$, namely $\boldsymbol{x}_* = (x_i)_{i=N_1+1}^{N_1+N_2}$, and wish to obtain an estimate of $f(x_i)$ for $i = N_1 + 1, \ldots, N_1 + N_2$. In the following, I write $\boldsymbol{y} = (y_1, \ldots, y_{N_1})^\top$ for the vector of training outcomes, $\overline{y} = (N_1)^{-1} \sum_{i=1}^{N_1} y_i$ for their sample mean, and $\boldsymbol{f}_* = (f(x_{N_1+1}), \ldots, f(x_{N_1+N_2}))^\top$ for the vector of test outcomes that we want to estimate.[1]

A positive-definite kernel function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ expresses the pairwise similarity between elements of $\mathcal{X}$. We construct the kernel matrix

$$\boldsymbol{K} := k(x_i, x_j)_{1 \leq i,j \leq N_1+N_2}$$

and define the submatrices

$$\begin{aligned}
\boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x}) &:= (K_{ij})_{1 \leq i,j \leq N_1}, \\
\boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x}_*) &:= (K_{ij})_{1 \leq i \leq N_1, N_1+1 \leq j \leq N_1+N_2}, \\
\boldsymbol{K}(\boldsymbol{x}_*, \boldsymbol{x}) &:= \boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x}_*)^\top, \\
\boldsymbol{K}(\boldsymbol{x}_*, \boldsymbol{x}_*) &:= (K_{ij})_{N_1+1 \leq i,j \leq N_1+N_2}.
\end{aligned}$$

In Gaussian process regression, the joint distribution of $\boldsymbol{y}$ and $\boldsymbol{f}_*$ is modelled as a multivariate normal distribution with covariance matrix

$$\begin{pmatrix} \boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1} & \boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x}_*) \\ \boldsymbol{K}(\boldsymbol{x}_*, \boldsymbol{x}) & \boldsymbol{K}(\boldsymbol{x}_*, \boldsymbol{x}_*) \end{pmatrix}.$$

The mean of this multivariate normal distribution is assumed to be zero. If the outcome data do not have a natural zero, it is standard practice to force this assumption by centring the outcomes around some $m \in \mathbb{R}$, with $m = \overline{y}$ being a common choice. In the formulae below, I made this

[1] The point predictions for a single test object x_{N_1+i} do not depend on the other components of $\boldsymbol{x}_*$. As long as the training objects $\boldsymbol{x}$ remain available so that the matrix $\boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x}_*)$ can be constructed, the test objects $\boldsymbol{x}_*$ do not need to be known at the time when the model's hyperparameters are tuned; see Section 3.3.

Data: $\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) \in \mathbb{R}^{N_1 \times N_1}, \boldsymbol{y} \in \mathbb{R}^{N_1}, \boldsymbol{K}(\boldsymbol{x}_*,\boldsymbol{x}) \in \mathbb{R}^{N_2 \times N_1}, m \in \mathbb{R}, \eta^2 \geq 0$
Result: Model predictions for $\boldsymbol{x}_*$

`GPR`$(\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}), \boldsymbol{y}, \boldsymbol{K}(\boldsymbol{x}_*,\boldsymbol{x}), m, \eta^2)$:
    $\boldsymbol{y} \leftarrow \boldsymbol{y} - m\mathbf{1}_{N_1}$
    $\boldsymbol{U} \leftarrow$ `cholesky`$(\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1})$ `/* i.e.,` $\boldsymbol{U}^\top \boldsymbol{U} = \boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1}$ `*/`
    $\boldsymbol{\alpha} \leftarrow \boldsymbol{U} \setminus (\boldsymbol{U}^\top \setminus \boldsymbol{y})$ `/* i.e.,` $\boldsymbol{\alpha} = (\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1})^{-1} \boldsymbol{y}$ `*/`
    $\overline{\boldsymbol{f}}_* \leftarrow \boldsymbol{K}(\boldsymbol{x}_*,\boldsymbol{x})\boldsymbol{\alpha} + m\mathbf{1}_{N_2}$
    return $\overline{\boldsymbol{f}}_*$

Algorithm 3.1: Obtaining predicted outcomes using Gaussian process regression.

centring step explicit:

$$\begin{bmatrix} \boldsymbol{y} - m\mathbf{1}_{N_1} \\ \boldsymbol{f}_* - m\mathbf{1}_{N_2} \end{bmatrix} \sim \mathcal{N}_{N_1+N_2}\left(\mathbf{0}, \begin{bmatrix} \boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1} & \boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}_*) \\ \boldsymbol{K}(\boldsymbol{x}_*,\boldsymbol{x}) & \boldsymbol{K}(\boldsymbol{x}_*,\boldsymbol{x}_*) \end{bmatrix}\right).$$

Having observed $\boldsymbol{x}, \boldsymbol{y}$ and $\boldsymbol{x}_*$, we can obtain the distribution of $\boldsymbol{f}_* - m\mathbf{1}_{N_2}$, which is multivariate normal, too:

$$(\boldsymbol{f}_* - m\mathbf{1}_{N_2})|\boldsymbol{x},\boldsymbol{y},\boldsymbol{x}_* \sim \mathcal{N}_{N_2}(\overline{\boldsymbol{f}}_*, \mathrm{Cov}(\boldsymbol{f}_*)),$$

where

$$\overline{\boldsymbol{f}}_* = \boldsymbol{K}(\boldsymbol{x}_*,\boldsymbol{x})(\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1})^{-1}(\boldsymbol{y} - m\mathbf{1}_{N_1}) + m\mathbf{1}_{N_2} \tag{3.1}$$

and

$$\mathrm{Cov}(\boldsymbol{f}_*) = \boldsymbol{K}(\boldsymbol{x}_*,\boldsymbol{x}_*) - \boldsymbol{K}(\boldsymbol{x}_*,\boldsymbol{x})(\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1})^{-1}\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}_*).$$

Equation 3.1 yields the model's point predictions; the diagonal elements of $\mathrm{Cov}(\boldsymbol{f}_*)$ can be used to construct credible intervals around these predictions.

Rasmussen & Williams (2006, Algorithm 2.1) provide an algorithm for efficiently estimating $\boldsymbol{f}_*$. I reproduce it here as Algorithm 3.1 with minor modifications. These modifications include implicitly incorporating the centring step, accommodating multiple test data points, and assuming that the Cholesky decomposition follows the convention in R and returns an upper triangular matrix rather than a lower triangular matrix.

Algorithm 3.1 assumes that $\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1}$ is positive-definite. In practical terms, a large enough value η^2 guarantees that this is the case as long as $\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x})$ is symmetric. Even if $\boldsymbol{K}(\boldsymbol{x},\boldsymbol{x})$ is positive-definite, slightly increasing the values on its main diagonal can help to avoid numerical issues with the Cholesky decomposition. The η^2 hyperparameter can be tuned, as explained later in this chapter.

3.2 Wasserstein and sliced Wasserstein kernels

As we saw in the previous section, the covariance of a Gaussian process is represented by a positive-definite kernel. I now turn to the matter of constructing such kernels based on Wasserstein and sliced Wasserstein distances. I restrict my attention to radial kernels, that is, kernels of the form

$$k(x, x') = \rho(d(x, x'))$$

for a distance d on $\mathcal{X}$ and a suitable function $\rho : [0, \infty) \to \mathbb{R}$.

Bachoc et al. (2020, Proposition 4) characterise fully the continuous functions $\rho : [0, \infty) \to \mathbb{R}$ that are suitable for constructing radial positive-definite kernels, provided that $\mathcal{X}$ is a Hilbert space such that $d(x, x') = \|x - x'\|_{\mathcal{X}}$. Here, $\|\cdot\|_{\mathcal{X}}$ is the norm induced by the inner product of the Hilbert space. In particular, the **power exponential function**

$$\rho_{s^2,t,p}(r) = s^2 \exp\left(-\left(\frac{r}{2t}\right)^p\right), \tag{3.2}$$

where $s^2, t > 0$ and $p \in (0, 2]$, serves as a suitable function ρ. Setting $p = 2$ in Equation 3.2 yields the **Gaussian radial basis function**. Another class of kernel functions is obtained by means of the **Matérn covariance function**

$$\rho_{s^2,t,\kappa}(r) = \frac{s^2 (r/t)^\kappa}{2^{\kappa-1}\Gamma(\kappa)} K_\kappa(r/t), \tag{3.3}$$

where $s^2, t, \kappa > 0$, where Γ is the Gamma function, and where K_κ is the modified Bessel function of the second kind. For $\kappa = 1/2, 3/2, 5/2$, Equation 3.3 simplifies to

$$\begin{aligned} \rho_{s^2,t,1/2}(r) &= s^2 \exp\left(-\frac{r}{2t}\right), \\ \rho_{s^2,t,3/2}(r) &= s^2 \left(1 + \frac{\sqrt{3}r}{2t}\right) \exp\left(-\frac{\sqrt{3}r}{2t}\right), \\ \text{and} \quad \rho_{s^2,t,5/2}(r) &= s^2 \left(1 + \frac{\sqrt{5}r}{2t} + \frac{5r^2}{12t^2}\right) \exp\left(-\frac{\sqrt{5}r}{2t}\right), \end{aligned}$$

respectively.

While Equations 3.2 and 3.3 allow us to define radial positive-definite kernels using the induced norm on a Hilbert space, the space of probability distributions with finite p-th moment is not a Hilbert space. In order to use Equations 3.2 and 3.3 when working with Wasserstein and sliced Wasserstein distances, then, we need to assess if these can be written in terms of an induced norm of a Hilbert space. This is where the notion of Hilbertian distances comes into play.

Definition 3.3 (Hilbertian distance)**.** Let $(\mathcal{X}, d)$ be a non-empty metric space. We say that the distance $d : \mathcal{X} \times \mathcal{X} \to [0, \infty)$ is **Hilbertian** if there exists a Hilbert space $\mathcal{H}$ and a mapping $\phi : \mathcal{X} \to \mathcal{H}$ such that

$$d(x, x') = \|\phi(x) - \phi(x')\|_{\mathcal{H}}$$

for any $x, x' \in \mathcal{X}$. Here, $\|\cdot\|_{\mathcal{H}} = \sqrt{\langle \cdot, \cdot \rangle_{\mathcal{H}}}$ is the norm induced by the inner product on $\mathcal{H}$. ◇

The idea is that if $d(x, x') = \|\phi(x) - \phi(x')\|_{\mathcal{H}}$ for some mapping $\phi : \mathcal{X} \to \mathcal{H}$, then we may define a radial kernel in terms of

$$k(x, x') = \rho(\|\phi(x) - \phi(x')\|_{\mathcal{H}})$$

using any of the suitable functions ρ identified by Bachoc et al. (2020).

Unfortunately, the p-Wasserstein distance on $\mathcal{P}_p(\mathbb{R}^d)$ is *not* Hilbertian for $d \geq 2$ and $p = 1, 2$. Peyré & Cuturi (2019, Proposition 8.2) show this by constructing a counterexample involving 35 mixtures of Dirac measures and demonstrating numerically that if the p-Wasserstein distance on $\mathcal{P}_p(\mathbb{R}^d), d \geq 2$, were Hilbertian, then this would contradict a result by Schoenberg (1938, Section 3). I refrain from reproducing the counterexample here. I could not find proofs pertaining to $p \notin \{1, 2\}$, but Peyré and Cuturi's construction generated counterexamples for all other (unspecified) values of p in the interval $[1, 4]$ that they considered, see their Figure 8.6. In any case, the values $p = 1, 2$ correspond to the most important use cases of Wasserstein distances.[2]

In the univariate case, however, a Hilbertian Wasserstein distance can be obtained as a corollary of Proposition 2.5 on page 11.

Corollary 3.4. The 2-Wasserstein distance is a Hilbertian distance on $\mathcal{P}_2(\mathbb{R})$. ♢

Proof. We denote by $L^2([0, 1])$ the quotient space obtained by identifying the functions in $\{f : [0, 1] \to \mathbb{R} : \int_0^1 |f(t)|^2 \mathrm{d}t < \infty\}$ agreeing Lebesgue-almost everywhere. When equipped with the inner product

$$\langle f, g \rangle_{L^2([0,1])} := \int_0^1 f(t)g(t)\mathrm{d}t,$$

$L^2([0, 1])$ is a Hilbert space. Its induced norm is

$$\|f\|_{L^2([0,1])} = \left(\int_0^1 f^2(t)\mathrm{d}t \right)^{\frac{1}{2}}.$$

We define the mapping $\phi : \mathcal{P}_2(\mathbb{R}) \to L^2([0, 1]), \mu \mapsto F_\mu^{-1}(\cdot)$, that associates each square-integrable univariate distribution to its quantile function. From Proposition 2.5, we now obtain

$$\mathbf{W}_2(\mu, \nu) = \left(\int_0^1 |\phi(\mu)(t) - \phi(\nu)(t)|^2 \mathrm{d}t \right)^{\frac{1}{2}} = \|\phi(\mu) - \phi(\nu)\|_{L^2([0,1])}. \qquad \square$$

Remark 3.5. In their Remark 2.30, Peyré & Cuturi (2019) claim that the p-Wasserstein is a Hilbertian distance on $\mathcal{P}_p(\mathbb{R})$ for all $p \geq 1$. Their remark hinges on the following argument:

> "[T]he Wasserstein distance is isometric to a linear space equipped with the L^p norm [for any $p \geq 1$, JV], or, equivalently, that the Wasserstein distance for measures on the real line is a Hilbertian metric."

The first part is readily seen by means of the mapping $\phi : \mathcal{P}_p(\mathbb{R}) \to L^p([0, 1]), \mu \mapsto F_\mu^{-1}(\cdot)$, as in the corollary above. But for $p \neq 2$, the L^p norm may not be induced by an inner product. By way of a counterexample, we define $f, g \in L^1([0, 1])$ as the indicator functions on $[0, 1/2]$ and on $(1/2, 1]$, respectively. If the L^1 norm were induced by an inner product, we would find

$$\|f + g\|^2_{L^1([0,1])} + \|f - g\|^2_{L^1([0,1])} = 2 \left(\|f\|^2_{L^1([0,1])} + \|g\|^2_{L^1([0,1])} \right).$$

[2] Incidentally, Bachoc et al. (2020) constructed positive-definite kernels on $\mathcal{P}_2(\mathbb{R}^d), d \geq 2$, by means of optimal transport maps rather than by means of the Wasserstein distances themselves.

However,

$$\|f+g\|^2_{L^1([0,1])} + \|f-g\|^2_{L^1([0,1])} = 2 \neq 1 = 2\left(\|f\|^2_{L^1([0,1])} + \|g\|^2_{L^1([0,1])}\right).$$

Consequently, the equivalence that Peyré & Cuturi (2019) rely on does not hold in general, and I will therefore not assume that the $\mathbf{W}_p$ distance is Hilbertian on $\mathcal{P}_p(\mathbb{R})$ for any $p \neq 2$. ♢

Because of Corollary 3.4, we may slot the $\mathbf{W}_2$ distance on $\mathcal{P}_2(\mathbb{R})$ into Equations 3.2 and 3.3 in order to obtain a positive-definite kernel function. Following Nguyen (2025, Chapter 4), I define the Wasserstein kernel for one-dimensional probability measures with finite second moment specifically in terms of a Gaussian radial basis function (i.e., setting p in Equation 3.2 to 2). That said, other kernel constructions would merit systematic exploration, too.

Definition 3.6 (Wasserstein kernel)**.** Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R})$ be one-dimensional probability measures with finite second moment. The **Wasserstein kernel** between μ and ν is defined as

$$k_W(\mu,\nu) := s^2 \exp\left\{-\frac{1}{2t^2}\mathbf{W}_2^2(\mu,\nu)\right\}$$

for a **scaling factor** $s^2 > 0$ and a **length-scale parameter** $t > 0$.[3] ♢

Kernels for the multivariate case can be constructed using sliced Wasserstein distances thanks to the following proposition (Nguyen, 2025, Proposition 4.10).

Proposition 3.7. The sliced Wasserstein distance of order $p = 2$ is Hilbertian. ♢

Proof. We denote by $L^2([0,1] \times \mathbb{S}^{d-1})$ the quotient space obtained by identifying the functions in

$$\left\{f : [0,1] \times \mathbb{S}^{d-1} \to \mathbb{R} : \int_{\mathbb{S}^{d-1}} \int_0^1 f^2(t,\theta)\mathrm{d}t\mathrm{d}\sigma(\theta) < \infty\right\}$$

agreeing Lebesgue-almost everywhere. When equipped with the inner product

$$\langle f, g\rangle_{L^2([0,1]\times\mathbb{S}^{d-1})} := \int_{\mathbb{S}^{d-1}} \int_0^1 f(t,\theta)g(t,\theta)\mathrm{d}t\mathrm{d}\sigma(\theta),$$

$L^2([0,1] \times \mathbb{S}^{d-1})$ is a Hilbert space, with the induced norm

$$\|f\|_{L^2([0,1]\times\mathbb{S}^{d-1})} = \left(\int_{\mathbb{S}^{d-1}} \int_0^1 f^2(t,\theta)\mathrm{d}t\mathrm{d}\sigma(\theta)\right)^{\frac{1}{2}} = \left(\mathbb{E}_{\boldsymbol{\theta}\sim\sigma}\left(\int_0^1 f^2(t,\boldsymbol{\theta})\mathrm{d}t\right)\right)^{\frac{1}{2}}.$$

[3] The scaling factor is usually denoted σ^2 in the literature, whereas the length-scale is more typically written as ℓ or reparametrised and written as γ. All of these symbols have other roles to play in this thesis, however.

We define the mapping $\phi : \mathcal{P}_2(\mathbb{R}^d) \to L^2([0,1] \times \mathbb{S}^{d-1})$ through $\phi(\mu)(t, \boldsymbol{\theta}) = F^{-1}_{\boldsymbol{\theta}^\star_\# \mu}(t)$. With Proposition 2.5, we obtain

$$\begin{aligned}
\mathbf{SW}_2(\mu, \nu) &= \left(\mathbb{E}_{\boldsymbol{\theta}\sim\sigma} \mathbf{W}_2^2(\boldsymbol{\theta}^\star_\# \mu, \boldsymbol{\theta}^\star_\# \nu)\right)^{\frac{1}{2}} \\
&= \left(\mathbb{E}_{\boldsymbol{\theta}\sim\sigma} \left(\int_0^1 \left|F^{-1}_{\boldsymbol{\theta}^\star_\# \mu}(t) - F^{-1}_{\boldsymbol{\theta}^\star_\# \nu}(t)\right|^2 \mathrm{d}t\right)\right)^{\frac{1}{2}} \\
&= \|\phi(\mu) - \phi(\nu)\|_{L^2([0,1]\times\mathbb{S}^{d-1})}.
\end{aligned}$$

□

Remark 3.8. Meunier et al. (2022) point out that the sliced Wasserstein distance of order $p = 1$ is not Hilbertian but prove that its square root *is* a Hilbertian distance (their Proposition 6). While I will follow most authors and focus on $p = 2$ in the remainder, Meunier et al.'s finding implies that we can slot $\sqrt{\mathbf{SW}_1(\cdot,\cdot)}$ into Equations 3.2 and 3.3 and obtain positive-definite kernels. ♢

Proposition 3.7 allows us to define the sliced Wasserstein kernel as follows.

Definition 3.9 (Sliced Wasserstein kernel)**.** Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ be d-dimensional probability measures with finite second moment, $d \geq 2$. The **sliced Wasserstein kernel** between μ and ν is defined as

$$k_{SW}(\mu, \nu) := s^2 \exp\left\{-\frac{1}{2t^2} \mathbf{SW}_2^2(\mu, \nu)\right\}$$

for a scaling factor $s^2 > 0$ and a length-scale parameter $t > 0$. ♢

In practice, $\mathbf{SW}_2^2(\mu, \nu)$ is estimated by Monte Carlo samples. Substituting $\widehat{\mathbf{SW}}_2^2(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L)$ for $\mathbf{SW}_2^2(\mu, \nu)$, we obtain

$$\hat{k}_{SW}(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L) = s^2 \exp\left\{-\frac{1}{2t^2} \widehat{\mathbf{SW}}_2^2(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L)\right\}.$$

For fixed projection directions, $\hat{k}_{SW}$ is a positive-definite kernel in its own right.

Proposition 3.10. For fixed $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L$, $\hat{k}_{SW}(\cdot, \cdot; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L)$ is a positive-definite kernel. ♢

Proof. For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d), d \geq 2$, we see that

$$\begin{aligned}
\hat{k}_{SW}(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L) &= s^2 \exp\left\{-\frac{1}{2t^2} \widehat{\mathbf{SW}}_2^2(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L)\right\} \\
&= s^2 \exp\left\{-\sum_{\ell=1}^{L} \frac{1}{2Lt^2} \mathbf{W}_2^2(\boldsymbol{\theta}^\star_{\ell\#} \mu, \boldsymbol{\theta}^\star_{\ell\#} \nu)\right\} \\
&= \prod_{\ell=1}^{L} s^{2/L} \exp\left\{-\frac{1}{2Lt^2} \mathbf{W}_2^2(\boldsymbol{\theta}^\star_{\ell\#} \mu, \boldsymbol{\theta}^\star_{\ell\#} \nu)\right\}.
\end{aligned}$$

The terms of this product are Wasserstein kernels with length-scale $\sqrt{L}t$ and with scaling factor $s^{2/L}$. Wasserstein kernels are positive-definite (Corollary 3.4), and products of positive-definite kernels are themselves positive-definite kernels (Lemma 3.2). □

However, as Nguyen (2025) points out, $\hat{k}_{SW}(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L)$ is a biased estimator of $k_{SW}(\mu, \nu)$: By Jensen's inequality, if $\mathrm{Var}_{\boldsymbol{\theta}\sim\sigma}\left(\mathbf{W}_2^2(\boldsymbol{\theta}_\#^\star\mu, \boldsymbol{\theta}_\#^\star\nu)\right) > 0$,

$$\mathbb{E}\left(\exp\left\{-\frac{1}{2t^2}\widehat{\mathbf{SW}}_2^2(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L)\right\}\right) > \exp\left\{-\frac{1}{2t^2}\mathbb{E}\left(\widehat{\mathbf{SW}}_2^2(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L)\right)\right\},$$

where the right-hand side equals $k_{SW}(\mu, \nu)$ with scaling factor $s^2 = 1$. Nguyen (2025) therefore defines the unbiased sliced Wasserstein kernel.

Definition 3.11 (Unbiased sliced Wasserstein kernel)**.** Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ be d-dimensional probability measures with finite second moment, $d \geq 2$. The **unbiased sliced Wasserstein kernel** between μ and ν is defined as

$$k_{USW}(\mu, \nu) := s^2\mathbb{E}_{\boldsymbol{\theta}\sim\sigma}\left(\exp\left\{-\frac{1}{2t^2}\mathbf{W}_2^2(\boldsymbol{\theta}_\#^\star\mu, \boldsymbol{\theta}_\#^\star\nu)\right\}\right)$$

for a scaling factor $s^2 > 0$ and a length-scale parameter $t > 0$. ♢

The unbiased sliced Wasserstein kernel is positive-definite (Nguyen, 2025, Proposition 4.11) and can be estimated in an unbiased way via Monte Carlo approximation:

$$\hat{k}_{USW}(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L) = \frac{s^2}{L}\sum_{\ell=1}^{L}\exp\left\{-\frac{1}{2t^2}\mathbf{W}_2^2(\boldsymbol{\theta}_{\ell\#}^\star\mu, \boldsymbol{\theta}_{\ell\#}^\star\nu)\right\},$$

for independent $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L \sim \sigma$.

To my knowledge, no study has carried out a direct comparison of the relative merits of $\hat{k}_{SW}$ and $\hat{k}_{USW}$ in terms of their usefulness for predictive modelling. In terms of computational demands, $\hat{k}_{USW}$ is necessarily somewhat more expensive to compute than $\hat{k}_{SW}$ as it requires evaluating L exponentials as opposed to merely one for each pair of distributions. More importantly, additional computational challenges arise when the application involves learning suitable values for s^2 and t from the data. For $\hat{k}_{USW}$, the kernel matrix must be reconstructed repeatedly during hyperparameter optimisation, and each reconstruction requires access to all L projected squared-distance matrices. By contrast, tuning s^2 and t for $\hat{k}_{SW}$ only depends on the single averaged squared-distance matrix, which is considerably cheaper to reuse in both time and memory. In my own attempts, I was not able to accelerate the computation of $\hat{k}_{USW}$ sufficiently to make it a viable candidate for hyperparameter tuning. I leave a focused comparison of $\hat{k}_{SW}$ and $\hat{k}_{USW}$ to future work.

Given a number of positive-definite kernels, Lemma 3.2 allows us to construct new kernels, for instance through summation.

Definition 3.12 (Sum of Wasserstein kernels)**.** Given two d-dimensional probability measures μ, ν and a set of vectors $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L \in \mathbb{R}^d$, we can construct the Wasserstein kernels

$$k_W^{(\ell)}(\boldsymbol{\theta}_{\ell\#}^\star\mu, \boldsymbol{\theta}_{\ell\#}^\star\nu) = s_\ell^2\exp\left\{-\frac{1}{2t_\ell^2}\mathbf{W}_2^2(\boldsymbol{\theta}_{\ell\#}^\star\mu, \boldsymbol{\theta}_{\ell\#}^\star\nu)\right\},$$

$\ell = 1, \ldots, L$. We define the function

$$k(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L) := \sum_{\ell=1}^{L} k_W^{(\ell)}(\boldsymbol{\theta}_{\ell\#}^{\star}\mu, \boldsymbol{\theta}_{\ell\#}^{\star}\nu)$$

with hyperparameters $t_1, \ldots, t_L, s_1^2, \ldots, s_L^2 > 0$, which is a positive-definite kernel in its own right. $\diamondsuit$

3.3 Hyperparameter tuning

When fitting a Gaussian process model using a (sliced) Wasserstein kernel, we need to decide on the values of at least three hyperparameters:

- the length-scale parameter $t > 0$ used in the exponential;
- the scaling factor $s^2 > 0$;
- and the noise variance parameter $\eta^2 > 0$ added to the main diagonal of $\boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x})$.

While it is possible to use cross-validation techniques when tuning these hyperparameters, an attractive feature of Gaussian process models is that the likelihood of the training data $\boldsymbol{y}$ given the predictor data $\boldsymbol{x}$ and the hyperparameters $\boldsymbol{\phi} = (t, s^2, \eta^2)^\top$ has a closed form. Specifically, the **negative log-marginal likelihood** (NLL) of a Gaussian process model is

$$-\log p(\tilde{\boldsymbol{y}}|\boldsymbol{x}, \boldsymbol{\phi}) = \frac{\tilde{\boldsymbol{y}}^\top \boldsymbol{\alpha}}{2} + \sum_{i=1}^{N_1} \log U_{ii} + \frac{N_1 \log(2\pi)}{2}, \tag{3.4}$$

where $\tilde{\boldsymbol{y}}$ is the possibly centred version of the outcome vector $\boldsymbol{y}$, $\boldsymbol{U}$ is the upper triangular matrix resulting from the Cholesky decomposition of $\boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1}$, and

$$\boldsymbol{\alpha} = (\boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1})^{-1} \tilde{\boldsymbol{y}} = \boldsymbol{U} \setminus (\boldsymbol{U}^\top \setminus \tilde{\boldsymbol{y}}).$$

This result is derived in Chapter 2 in Rasmussen & Williams (2006, see their Equation 2.30). The expression is obtained by marginalising over the unknown values of $\boldsymbol{f}_*$, hence its name.

To find the hyperparameter values minimising the NLL, I use the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm (see Nocedal & Wright, 2006, Chapter 6). While this algorithm can estimate gradients on the fly using the finite-differences method, the gradient of the NLL with respect to $\boldsymbol{\phi}$ is not too arduous to compute explicitly. In order to enforce positive hyperparameter values, I optimise the NLL with respect to $\log(t), \log(s^2)$ and $\log(\eta^2)$ instead of with respect to t, s^2 and η^2 directly.

From Rasmussen & Williams (2006, Equation 5.9), we obtain the k-th partial derivative of the NLL as

$$-\frac{\partial}{\partial \phi_k} \log p(\tilde{\boldsymbol{y}}|\boldsymbol{x}, \boldsymbol{\phi}) = -\frac{1}{2} \operatorname{Tr}\left((\boldsymbol{\alpha}\boldsymbol{\alpha}^\top - \boldsymbol{K}^{-1}) \frac{\partial \boldsymbol{K}}{\partial \phi_k} \right),$$

where

$$\boldsymbol{K} = \boldsymbol{K}(\boldsymbol{x}, \boldsymbol{x}) + \eta^2 \boldsymbol{I}_{N_1}$$

and

$$\boldsymbol{\alpha} = \boldsymbol{K}^{-1}\tilde{\boldsymbol{y}}.$$

It remains to compute $\partial \boldsymbol{K}/\partial \phi_k$ for $k = 1, 2, 3$. Writing $d^2(x_i, x_j)$ for the squared distance between the input objects x_i and x_j, the partial derivative for the log length-scale parameter is

$$\begin{aligned}
\left(\frac{\partial \boldsymbol{K}}{\partial \log t}\right)_{ij} &= \frac{\partial}{\partial \log(t)} \left(s^2 \exp\left\{ -\frac{d^2(x_i, x_j)}{2\exp(2\log(t))} \right\} + \eta^2[i = j] \right) \\
&= \frac{s^2 d^2(x_i, x_j)}{t^2} \exp\left\{ -\frac{d^2(x_i, x_j)}{2t^2} \right\}.
\end{aligned}$$

For the log scaling factor, we obtain

$$\begin{aligned}
\left(\frac{\partial \boldsymbol{K}}{\partial \log s^2}\right)_{ij} &= \frac{\partial}{\partial \log(s^2)} \left(\exp(\log(s^2)) \exp\left\{ -\frac{d^2(x_i, x_j)}{2t^2} \right\} + \eta^2[i = j] \right) \\
&= s^2 \exp\left\{ -\frac{d^2(x_i, x_j)}{2t^2} \right\}.
\end{aligned}$$

For the noise variance parameter, finally, we get

$$\begin{aligned}
\left(\frac{\partial \boldsymbol{K}}{\partial \log \eta^2}\right)_{ij} &= \frac{\partial}{\partial \log(\eta^2)} \left(s^2 \exp\left\{ -\frac{d^2(x_i, x_j)}{2t^2} \right\} + \exp(\log \eta^2)[i = j] \right) \\
&= \eta^2[i = j].
\end{aligned}$$

For the initialisation, I generated independent starting parameter values as follows:

$$\begin{aligned}
\log(t) &\sim \mathrm{Uniform}(\log(0.1 \cdot \mathrm{MedDist}), \log(10 \cdot \mathrm{MedDist})), \\
\log(s^2) &\sim \mathrm{Uniform}(\log 0.1, \log 10), \\
\log(\eta^2) &\sim \mathrm{Uniform}(-10, -2),
\end{aligned}$$

where

$$\mathrm{MedDist} = \mathrm{median}(d(x_i, x_j))_{1 \le i < j \le N_1}.$$

To prevent the BFGS algorithm from being stuck near a local minimum, the search can be run from multiple independent starting points.

Remark 3.13 (Hyperparameter tuning for sums of kernels)**.** In Definition 3.12, I introduced a kernel defined as the sum of L Wasserstein kernels:

$$k(\mu, \nu; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L) = \sum_{\ell=1}^{L} k_W^{(\ell)}(\boldsymbol{\theta}_{\ell\#}^{\star}\mu, \boldsymbol{\theta}_{\ell\#}^{\star}\nu)$$

with hyperparameters $t_1, \ldots, t_L, s_1^2, \ldots, s_L^2 > 0$. Adding the noise variance parameter from the Gaussian process model, this yields $2L + 1$ hyperparameters that can be tuned in the same fashion as in the single-kernel case discussed above, albeit at greater computational expense.

Specifically, the partial derivative of K_{ij} with respect to $\log t_\ell$ is

$$\left(\frac{\partial \boldsymbol{K}}{\partial \log t_\ell}\right)_{ij} = \frac{s_\ell^2 d_\ell^2(x_i, x_j)}{t_\ell^2} \exp\left\{-\frac{d_\ell^2(x_i, x_j)}{2t_\ell^2}\right\},$$

and the partial derivative of K_{ij} with respect to $\log s_\ell^2$ is

$$\left(\frac{\partial \boldsymbol{K}}{\partial \log s_\ell^2}\right)_{ij} = s_\ell^2 \exp\left\{-\frac{d_\ell^2(x_i, x_j)}{2t_\ell^2}\right\},$$

where $d_\ell^2(x_i, x_j)$ denotes the squared distance between the ℓ-th projections of x_i and x_j, $\ell = 1, \ldots, L$. The partial derivative of K_{ij} with respect to $\log \eta^2$ is the same as before, $\eta^2[i = j]$. $\diamondsuit$

Chapter 4

A toy example

Chapter 2 introduced Wasserstein and sliced Wasserstein distances. Chapter 3 showed how kernels can be derived from these distances and explained how these kernels can be used in predictive modelling, specifically in Gaussian process regression. In this chapter, I illustrate how these tools can be put into practice. I do so by means of a toy example constructed specially to demonstrate the usefulness of sliced Wasserstein kernels in a situation where both standard vector-based approaches and Wasserstein kernels fail abjectly.

4.1 Set-up

Input distributions. For each $a > 0$, we define the bivariate Gaussian mixture distribution

$$P(a) := \frac{1}{4}\sum_{i=1}^{4} \mathcal{N}_2(\boldsymbol{\mu}_i, \boldsymbol{I}_2),$$

where

$$\boldsymbol{\mu}_1 = \begin{pmatrix} a \\ 0 \end{pmatrix}, \quad \boldsymbol{\mu}_2 = \begin{pmatrix} 0 \\ a \end{pmatrix}, \quad \boldsymbol{\mu}_3 = -\boldsymbol{\mu}_1, \quad \boldsymbol{\mu}_4 = -\boldsymbol{\mu}_2.$$

For each $\omega \in \mathbb{R}$, we further define

$$P(a, \omega) := \mathcal{L}(\boldsymbol{R}(\omega)\boldsymbol{X}),$$

where

$$\boldsymbol{R}(\omega) = \begin{pmatrix} \cos\omega & -\sin\omega \\ \sin\omega & \cos\omega \end{pmatrix}$$

and $\boldsymbol{X} \sim P(a)$. That is, $P(a, \omega)$ takes the Gaussian mixture $P(a)$ and rotates it counterclockwise by an angle of ω; see Figure 4.1 on the next page. For $\boldsymbol{Z} \sim \mathcal{N}_2(\boldsymbol{\mu}_i, \boldsymbol{I}_2)$, the random vector $\boldsymbol{R}(\pi/2)\boldsymbol{Z}$ has the distribution $\mathcal{N}_2(\boldsymbol{\mu}_{i+1 \bmod 4}, \boldsymbol{I}_2), i = 1, \ldots, 4$. The angle $\pi/2$ is the smallest positive angle for which this is true. As a result, the period of $\omega \mapsto P(a, \omega)$ is $\pi/2$.

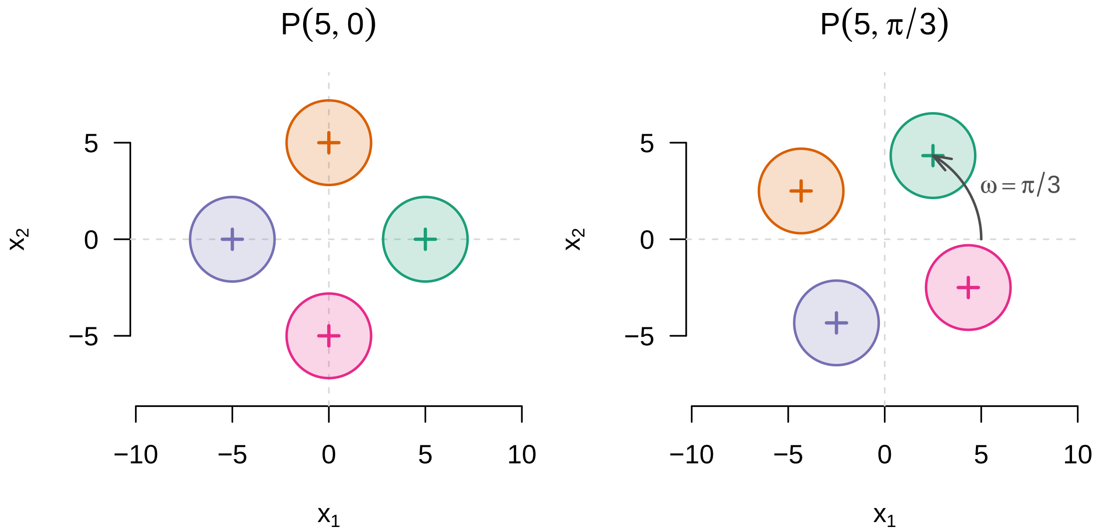

Figure 4.1: For this toy example, the data are generated from Gaussian mixture distributions whose components are laid out symmetrically around the origin. The a parameter (here $a = 5$) governs the distance of the component centroids from the origin; the ω parameter ($\omega = \pi/3$ in the right panel) determines the rotation applied to the unrotated mixture $P(a, 0)$. The ellipses represent 95% probability contours of each mixture component, the centroids of which are marked by crosses.

For any a, ω, the expectation of $\boldsymbol{X} \sim P(a, \omega)$ is the zero vector in $\mathbb{R}^2$. Further, the distribution $P(a, 0)$ is symmetrical about both cardinal axes. Hence, for $\boldsymbol{X} = (X_1, X_2) \sim P(a, 0)$,

$$(X_1, X_2) \stackrel{d}{=} (X_1, -X_2),$$

so

$$\mathrm{Cov}(X_1, X_2) = \mathrm{Cov}(X_1, -X_2) = -\mathrm{Cov}(X_1, X_2),$$

which means that $\mathrm{Cov}(\boldsymbol{X})$ is diagonal. Since the first and second component are exchangeable, the diagonal entries of the covariance matrix are identical. Along one axis, the variance of the mixture is the sum of within- and between-cluster variances, i.e., $a^2 + 1$. Hence, the covariance matrix of $P(a, 0)$ is $(a^2 + 1)\boldsymbol{I}_2$. The covariance matrix of $P(a, \omega)$ has the same form:

$$\boldsymbol{R}(\omega)(a^2 + 1)\boldsymbol{I}_2\boldsymbol{R}(\omega)^\top = (a^2 + 1)\boldsymbol{R}(\omega)\boldsymbol{R}(\omega)^\top = (a^2 + 1)\boldsymbol{I}_2.$$

The first and second moments of $P(a, \omega)$, then, do not depend on ω, and the first moment does not depend on a, either. Consequently, any vector-based model that uses these moments will not be useful for predicting the rotation angle ω in test data.

Let us now turn our attention to the marginal distributions of $(X_1, X_2) \sim P(a, \pi/4 + \omega)$. The projection of the first mixture component onto the x-axis is a normal distribution with expectation $a\cos(\pi/4 + \omega)$ and unit variance; the projection of the fourth mixture component onto the x-axis is again normal with unit variance, but with expectation $a\sin(\pi/4 + \omega)$. By symmetry, we find that the distribution of X_1 is

$$\mathcal{L}(X_1) = \frac{1}{4}\Big(\mathcal{N}(a\cos(\pi/4+\omega),1) + \mathcal{N}(-a\sin(\pi/4+\omega),1) + \mathcal{N}(-a\cos(\pi/4+\omega),1) + \mathcal{N}(a\sin(\pi/4+\omega),1)\Big).$$

Similarly, for $(Y_1, Y_2) \sim P(a, \pi/4-\omega)$, we obtain

$$\begin{aligned}\mathcal{L}(Y_1) &= \frac{1}{4}\Big(\mathcal{N}(a\cos(\pi/4-\omega),1) + \mathcal{N}(-a\sin(\pi/4-\omega),1) + \\ &\qquad \mathcal{N}(-a\cos(\pi/4-\omega),1) + \mathcal{N}(a\sin(\pi/4-\omega),1)\Big) \\ &= \frac{1}{4}\Big(\mathcal{N}(a\sin(\pi/4+\omega),1) + \mathcal{N}(-a\cos(\pi/4+\omega),1) + \\ &\qquad \mathcal{N}(-a\sin(\pi/4+\omega),1) + \mathcal{N}(-a\cos(\pi/4+\omega),1)\Big) \\ &= \mathcal{L}(X_1).\end{aligned}$$

By the same token, $\mathcal{L}(X_2) = \mathcal{L}(Y_2)$. Thus, any method relying solely on the marginal distributions of the input distributions will not be able to tell $P(a, \pi/4+\omega)$ and $P(a, \pi/4-\omega)$ apart.

Scenarios. We consider an artificial setting in which we observe empirical distributions $\widehat{P}_1, \ldots, \widehat{P}_{N_1+N_2}$ obtained from simple random samples from the distributions $P(a, \omega_1), \ldots, P(a, \omega_{N_1+N_2})$. The rotation angles $\omega_1, \ldots, \omega_{N_1}$ with which the training data were generated are known; the objective is to retrieve the values of $\omega_{N_1+1}, \ldots, \omega_{N_1+N_2}$ that were used to generate the test set, which are unknown. Each sample contains the same number of observations, n.

The utility of sliced Wasserstein kernels will be illustrated in two scenarios.

- In Scenario 1, the rotation angles are drawn independently from a uniform distribution over the interval $(0, \pi/4)$. Over this interval, the images of cos and sin do not overlap. As a result, the mappings from the rotation angle to the marginal distributions of $P(a, \omega)$ are injective over this fairly narrow interval, and Wasserstein kernels are expected to perform relatively well.

- In Scenario 2, the rotation angles are drawn from a uniform distribution over the interval $(0, \pi/2)$. While the mapping $\omega \mapsto P(a, \omega)$ is injective over this wider interval, the mappings from the rotation angle to the marginal distributions of $P(a, \omega)$ are not. Consequently, Wasserstein kernels are expected to show degraded performance, whereas sliced Wasserstein kernels, which take into account the whole distribution, are still expected to retain some predictive power.

In both scenarios, I set $a = 5, N_1 = N_2 = 50$ and $n = 200$. The values of $\omega_1, \ldots, \omega_{N_1+N_2}$ were sampled independently from Uniform$(0, \pi/4)$ (Scenario 1) and Uniform$(0, \pi/2)$ (Scenario 2). For both scenarios, twenty independent realisations of the rotation angles and the corresponding samples are generated. Each of the resulting twenty data sets per scenario is analysed independently of the others.

4.2 Method of analysis

For each of the twenty data sets per scenario, the closeness of the predicted outcomes $\widehat{\omega}_{N_1+1}, \ldots, \widehat{\omega}_{N_1+N_2}$ to the test data $\omega_{N_1+1}, \ldots, \omega_{N_1+N_2}$ is assessed quantitatively by means of the root mean squared error (RMSE)

$$\text{RMSE} := \sqrt{\frac{1}{N_2} \sum_{i=N_1+1}^{N_1+N_2} (\widehat{\omega}_i - \omega_i)^2}.$$

The range of the outcomes varies between the two scenarios. In order to present the results in a more directly comparable fashion, an R^2 value is computed as

$$R^2 := 1 - \frac{\sum_{i=N_1+1}^{N_1+N_2} (\widehat{\omega}_i - \omega_i)^2}{\sum_{i=N_1+1}^{N_1+N_2} (\overline{\omega} - \omega_i)^2},$$

where $\overline{\omega} = N_1^{-1} \sum_{j=1}^{N_1} \omega_j$ is the mean outcome in the training data. Higher R^2 values correspond to greater predictive accuracy. This definition allows R^2 to be negative when the predictions are worse than when simply using the training set mean.

Baselines. A reference baseline is established by setting $\widehat{\omega}_i$ to $\overline{\omega} = N_1^{-1} \sum_{j=1}^{N_1} \omega_j$, the mean outcome in the training set, $i = N_1 + 1, \ldots, N_1 + N_2$. By definition, the R^2 value for this reference baseline is 0.

While we have established that there is no signal of the outcome in the means and covariances of the input distributions, I nonetheless applied a vector-based method to this regression problem in which each input distribution was summarised by the sample means and sample covariances of its first and second dimensions as well as by the Pearson correlation between its first and second dimensions. For each distribution, this five-dimensional vector was used as a predictor in a random forest model (see Breiman, 2001). This model's `mtry` hyperparameter was tuned by minimising the out-of-bag mean squared error over the set $\{1, \ldots, 5\}$. The `ntree` hyperparameter, which rarely substantially affects the performance of random forests, was set to 500.

Wasserstein kernels. Next, a number of Gaussian process models were fitted to the data. The first two of these models were based on Wasserstein kernels (see Definition 3.6 on page 21) for which the inputs were the first and second marginals of the empirical distributions. That is, for each pair of empirical input distributions $\widehat{P}(a, \omega_i), \widehat{P}(a, \omega_j)$, I computed

$$k_W(\boldsymbol{e}_{1\#}^{\star}\widehat{P}(a, \omega_i), \boldsymbol{e}_{1\#}^{\star}\widehat{P}(a, \omega_j))$$

and

$$k_W(\boldsymbol{e}_{2\#}^{\star}\widehat{P}(a, \omega_i), \boldsymbol{e}_{2\#}^{\star}\widehat{P}(a, \omega_j)),$$

where $\boldsymbol{e}_1 = (1, 0)^\top$ and $\boldsymbol{e}_2 = (0, 1)^\top$.

The hyperparameters s^2, t and η^2 were tuned by minimising the negative log-marginal likelihood using the BFGS algorithm as described in Section 3.3. The hyperparameter search was initialised ten different times in order to prevent the optimisation algorithm from being stuck near a local minimum.

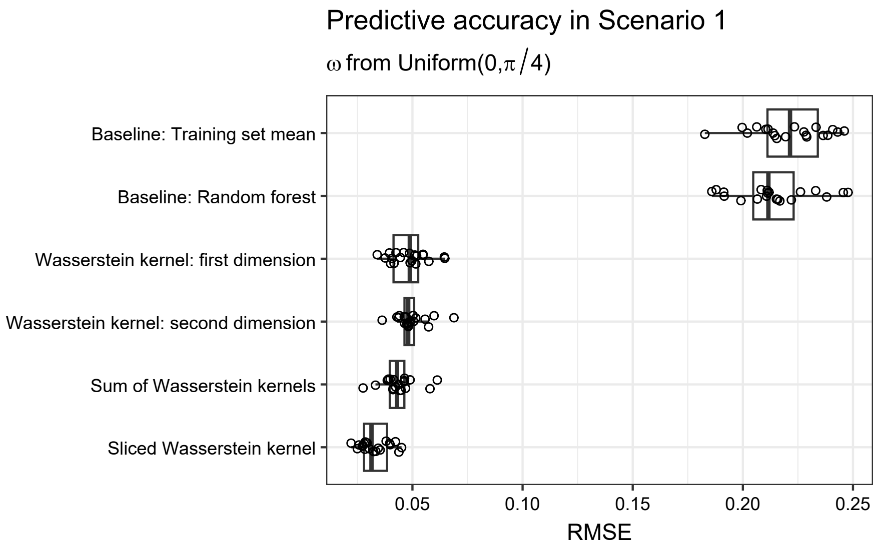

Figure 4.2: The root mean squared errors obtained by each prediction model for the twenty runs of the toy example in the first scenario (rotation angle drawn from Uniform$(0, \pi/4)$).

Sum of Wasserstein kernels. For each pair of empirical input distributions $\widehat{P}(a, \omega_i), \widehat{P}(a, \omega_j)$, I computed the kernel value

$$k(\widehat{P}(a, \omega_i), \widehat{P}(a, \omega_j); \boldsymbol{e}_1, \boldsymbol{e}_2),$$

see Definition 3.12 on page 23. This kernel combines the information from both marginals into a single kernel, potentially allowing the model to use both projection directions simultaneously. The kernel was used in another Gaussian process model, whose hyperparameters s_1^2, s_2^2, t_1, t_2 and η^2 were tuned as explained in Remark 3.13.

Sliced Wasserstein kernel. Finally, a Gaussian process model using the sliced Wasserstein kernel (Definition 3.9 on page 22) was fitted to the data. This kernel was estimated using Monte Carlo simulation based on $L = 50$ projection directions sampled i.i.d. from a uniform distribution over $\mathbb{S}^1$. For each of the twenty data sets, the projection directions were sampled anew. The hyperparameters s^2, t and η^2 were again tuned by minimising the negative log-marginal likelihood, and, again, the BFGS algorithm was initialised ten different times for each hyperparameter search.

4.3 Results

Table 4.1 and Figure 4.2 show the RMSE values achieved by each modelling approach in the scenario in which the rotation angles were sampled from a uniform distribution over the interval $(0, \pi/4)$. As expected, the three Wasserstein kernels substantially outperform both the naïve baseline and the random forest baseline. However, they are themselves outperformed by the sliced Wasserstein kernel. This suggests that the sliced Wasserstein kernel picks up on, and is able to leverage, discriminatory information that is not contained in the input distributions' marginals.

Table 4.1: The root mean squared error obtained by different prediction models over twenty runs of the toy example in the first scenario (rotation angle drawn from Uniform$(0, \pi/4)$).

Method of analysis	Mean	Standard deviation
Baseline: Training set mean	0.221	0.017
Baseline: Random forest	0.214	0.018
Wasserstein kernel: first dimension	0.048	0.008
Wasserstein kernel: second dimension	0.049	0.007
Sum of Wasserstein kernels	0.044	0.007
Sliced Wasserstein kernel	0.033	0.007

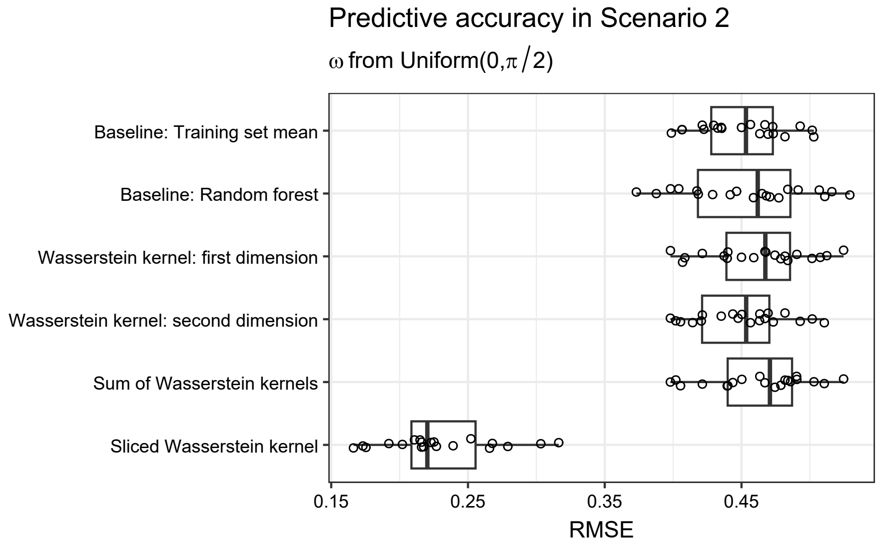

Figure 4.3: The root mean squared errors obtained by each prediction model for the twenty runs of the toy example in the second scenario (rotation angles drawn from Uniform$(0, \pi/2)$).

In the second scenario, in which the rotation angles are drawn from a uniform distribution over the interval $(0, \pi/2)$, the advantage of the sliced Wasserstein kernel relative to the other approaches is even more pronounced, as expected; see Table 4.2 and Figure 4.3. Indeed, the three Wasserstein kernels do not outperform the baselines. The sliced Wasserstein kernel, by contrast, is clearly able to pick up on some signal in the input distributions that is not restricted to the marginals.

The R^2 values, shown in Table 4.3 and Figure 4.4, tell the same story: in Scenario 1, the Wasserstein kernels substantially outperform their baselines but they are narrowly outperformed by the sliced Wasserstein kernel; in Scenario 2, the Wasserstein kernels cannot be used to reliably predict the outcome, whereas the sliced Wasserstein kernel does a reasonable job, achieving a mean R^2 of 0.74.

Table 4.2: The root mean squared error obtained by different prediction models over twenty runs of the toy example in the second scenario (rotation angles drawn from Uniform$(0, \pi/2)$).

Method of analysis	Mean	Standard deviation
Baseline: Training set mean	0.451	0.032
Baseline: Random forest	0.455	0.045
Wasserstein kernel: first dimension	0.463	0.037
Wasserstein kernel: second dimension	0.451	0.033
Sum of Wasserstein kernels	0.463	0.037
Sliced Wasserstein kernel	0.229	0.041

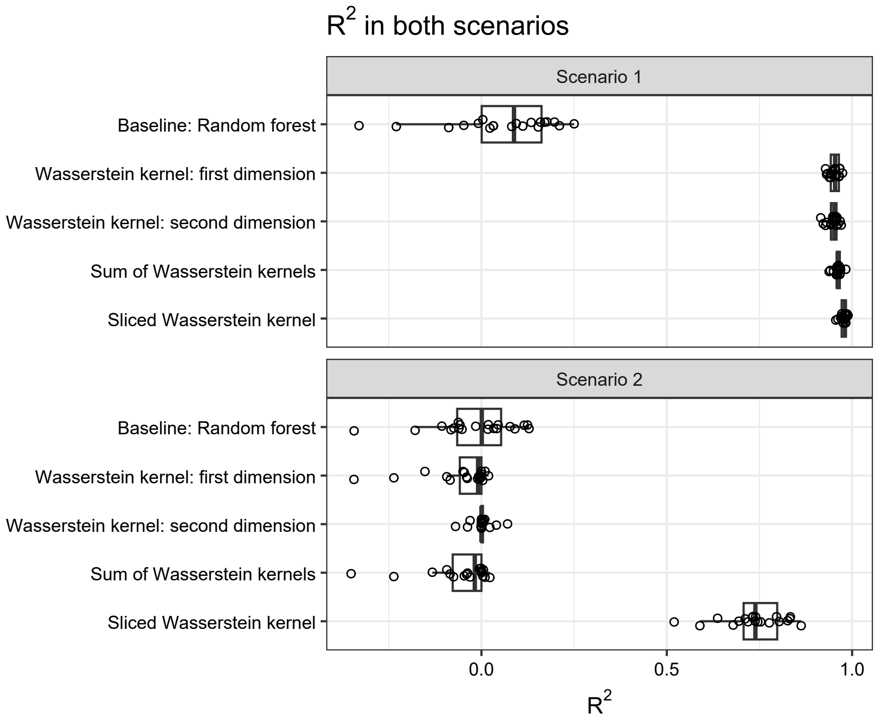

Figure 4.4: The R^2 values obtained by each predictive model for the twenty runs of the toy example.

Table 4.3: The mean and standard deviation of the R^2 values obtained by different prediction models over twenty runs of the toy example.

Method of analysis	Scenario 1	Scenario 2
Baseline: Random forest	0.06 $\pm$ 0.15	-0.02 $\pm$ 0.11
Wasserstein kernel: first dimension	0.95 $\pm$ 0.01	-0.05 $\pm$ 0.09
Wasserstein kernel: second dimension	0.95 $\pm$ 0.01	0.00 $\pm$ 0.03
Sum of Wasserstein kernels	0.96 $\pm$ 0.01	-0.05 $\pm$ 0.09
Sliced Wasserstein kernel	0.98 $\pm$ 0.01	0.74 $\pm$ 0.08

4.4 Conclusion

The results presented in the previous section demonstrate that sliced Wasserstein kernels can perform well where vector-based methods and kernels based on marginal distributions struggle. The toy example, of course, was constructed specially for this purpose. However, the performance of any kernel-based method is likely to depend on how the input distributions are scaled and oriented in the ambient space. Unlike in the present toy example, then, a real-life application of Wasserstein and sliced Wasserstein kernels will likely involve some preprocessing of the input distributions in order to find a suitable scaling and orientation. In order to provide a theoretical basis for some of the preprocessing choices that naturally arise in practice, the following chapter studies the effect of global affine transformations of the input distribution on sliced Wasserstein distances and the resultant kernels.

Chapter 5

Effects of global affine transformations

The previous chapter illustrated the sliced Wasserstein–Gaussian process regression workflow as well as its potential usefulness for tackling regression problems in which the inputs are probability distributions. The toy example used, however, was deliberately artificial, which allowed me to disregard the difficulties that analysts are likely to face in real-life applications.

One source of difficulty is that the variables that make up the input distributions are typically measured in arbitrary units. For example, if the input distributions consists of measurements of body heights and masses, the sliced Wasserstein distances between two distributions will depend on whether the heights are expressed in metres, centimetres or inches and on whether the masses are expressed in kilograms or pounds. Rescaling a single variable can drastically alter the pairwise sliced Wasserstein distances, which in turn may affect the performance of the prediction model downstream. Indeed, the asymptotic analyses in this chapter will show that such rescaling may render the resulting kernel matrix degenerate.

A popular approach for dealing with arbitrary measurement units is to centre and scale each input distribution using pooled estimates of the variables' means and standard deviations. That is, if $\boldsymbol{M}_i$ is the $n_i \times d$ matrix representing the i-th input distribution, we compute

$$\widetilde{\boldsymbol{M}_i} = (\boldsymbol{M}_i - \mathbf{1}_{n_i}\widehat{\boldsymbol{\mu}}^\top) \begin{pmatrix} \widehat{\sigma}_1^{-1} & & \\ & \ddots & \\ & & \widehat{\sigma}_d^{-1} \end{pmatrix},$$

where $\widehat{\boldsymbol{\mu}} \in \mathbb{R}^d$ is a pooled estimate of the mean and where $\widehat{\sigma}_1, \dots, \widehat{\sigma}_d$ are pooled estimates of the standard deviations.

An alternative approach is to 'whiten' the input distributions. The goal of whitening is to transform the input in such a way that the pooled sample covariance matrix of the transformed input is the identity matrix. In our context, this amounts to computing

$$\widetilde{\boldsymbol{M}_i} = \boldsymbol{M}_i\boldsymbol{W}.$$

Here, $\boldsymbol{W} \in \mathbb{R}^{d\times d}$ is a matrix such that $\boldsymbol{W}\widehat{\boldsymbol{\Sigma}}\boldsymbol{W}^\top = \boldsymbol{I}_d$, where $\widehat{\boldsymbol{\Sigma}}$ is a pooled estimate of the covariance matrix of the input distributions. Equivalently, $\boldsymbol{W}$ has the property that $\boldsymbol{W}^\top\boldsymbol{W} = \widehat{\boldsymbol{\Sigma}}^{-1}$.

An infinite number of matrices $\boldsymbol{W}$ fulfilling this criterion exist for any given estimated covariance matrix, but a handful of procedures for constructing one are used in practice. These include Mahalanobis (or ZCA) whitening, Cholesky whitening, and PCA whitening (see Kessy et al., 2018, for an overview). I will discuss these different whitening techniques in more detail later.

Both scaling and whitening are instances of a global affine transformation: the same transformation is applied to every input distribution, as opposed to applying different transformations to different input distributions. Such latter (local) transformations may nonetheless be useful. For instance, it is conceivable that analysts would want to isolate shape differences between the input distributions from location and scale differences. Scaling and centring each distribution separately prior to computing the distances would accomplish this. The goal of this chapter, however, is to better understand the effects of such global affine transformations on sliced Wasserstein distances and the kernels derived from them. It will show that such transformations amount to a distortion of the measure on the unit sphere and will derive the Radon–Nikodym derivative of this distortion. The theoretical findings are illustrated by means of simulations in which the degree of anisotropy of the transformation matrix is varied systematically. The chapter ends with a real-data example that demonstrates the effects of global scaling and whitening in a less artificial setting.

5.1 Theoretical considerations

Let $P, Q \in \mathcal{P}_2(\mathbb{R}^d), d \geq 1$, and let $\boldsymbol{X} \sim P, \boldsymbol{Y} \sim Q$. Further let $\boldsymbol{A} \in \mathrm{GL}_d(\mathbb{R})$ be an invertible matrix and $\boldsymbol{b} \in \mathbb{R}^d$ a translation vector. For $\boldsymbol{\theta} \in \mathbb{R}^d$, we have

$$\langle \boldsymbol{AX} + \boldsymbol{b}, \boldsymbol{\theta} \rangle = \langle \boldsymbol{X}, \boldsymbol{A}^\top \boldsymbol{\theta} \rangle + \langle \boldsymbol{b}, \boldsymbol{\theta} \rangle.$$

Consequently, the squared sliced Wasserstein distance of order 2 between the distributions of $\boldsymbol{AX} + \boldsymbol{b}$ and $\boldsymbol{AY} + \boldsymbol{b}$ is

$$\begin{aligned}\mathbf{SW}_2^2(\boldsymbol{AX}+\boldsymbol{b}, \boldsymbol{AY}+\boldsymbol{b}) &= \int \mathbf{W}_2^2(\langle \boldsymbol{AX}+\boldsymbol{b}, \theta\rangle, \langle \boldsymbol{AY}+\boldsymbol{b}, \theta\rangle)\mathrm{d}\sigma(\theta) \\ &= \int \mathbf{W}_2^2(\langle \boldsymbol{X}, \boldsymbol{A}^\top\theta\rangle + \langle \boldsymbol{b}, \theta\rangle, \langle \boldsymbol{Y}, \boldsymbol{A}^\top\theta\rangle + \langle \boldsymbol{b}, \theta\rangle)\mathrm{d}\sigma(\theta).\end{aligned}$$

Since the Wasserstein distance is translation-invariant (see Lemma 2.2 on page 8), this simplifies to

$$\mathbf{SW}_2^2(\boldsymbol{AX}+\boldsymbol{b}, \boldsymbol{AY}+\boldsymbol{b}) = \int \mathbf{W}_2^2(\langle \boldsymbol{X}, \boldsymbol{A}^\top\theta\rangle, \langle \boldsymbol{Y}, \boldsymbol{A}^\top\theta\rangle)\mathrm{d}\sigma(\theta) = \mathbf{SW}_2^2(\boldsymbol{AX}, \boldsymbol{AY}). \tag{5.1}$$

Remark 5.1 (Change of measure)**.** By definition,

$$\mathbf{SW}_2^2(\boldsymbol{X}, \boldsymbol{Y}) = \int \mathbf{W}_2^2(\langle \boldsymbol{X}, \theta\rangle, \langle \boldsymbol{Y}, \theta\rangle)\mathrm{d}\sigma(\theta).$$

Comparing this expression to Equation 5.1, we observe that

$$\mathbf{SW}_2^2(\boldsymbol{AX}+\boldsymbol{b}, \boldsymbol{AY}+\boldsymbol{b}) = \int \mathbf{W}_2^2(\langle \boldsymbol{X}, \theta\rangle, \langle \boldsymbol{Y}, \theta\rangle)\mathrm{d}\sigma_{\boldsymbol{A}}(\theta),$$

with $\sigma_{\boldsymbol{A}}$ the pushforward measure of σ under the mapping $\mathbb{R}^d \ni \boldsymbol{v} \mapsto \boldsymbol{A}^\top \boldsymbol{v} \in \mathbb{R}^d$. Global affine transformations, then, act on sliced Wasserstein distances solely through a distortion of the measure on $\mathbb{S}^{d-1}$. ♢

To better characterise this change of measure, we write $\mathbf{SW}_2^2(\boldsymbol{X}, \boldsymbol{Y})$ and $\mathbf{SW}_2^2(\boldsymbol{AX}, \boldsymbol{AY})$ as integrals with respect to the Lebesgue measure. For $\boldsymbol{Z} \sim \mathcal{N}_d(\boldsymbol{0}, \boldsymbol{I}_d)$, we have $\mathbb{P}(\|Z\| = 0) = 0$ and it holds that $\boldsymbol{Z}/\|\boldsymbol{Z}\| \sim \text{Uniform}(\mathbb{S}^{d-1})$. Hence,

$$\mathbf{SW}_2^2(\boldsymbol{X}, \boldsymbol{Y}) = \int_{\mathbb{R}^d} \mathbf{W}_2^2(\langle \boldsymbol{X}, z/\|z\|\rangle, \langle \boldsymbol{Y}, z/\|z\|\rangle)\varphi(z)\mathrm{d}z,$$

where $\varphi : \mathbb{R}^d \to (0, \infty), \boldsymbol{z} \mapsto (2\pi)^{-d/2} \exp\left(-\|\boldsymbol{z}\|^2/2\right)$, is the canonical Lebesgue density of the d-variate standard normal distribution. Similarly, and after using the homogeneity property of the Wasserstein distance (Lemma 2.2), we see that

$$\begin{aligned}\mathbf{SW}_2^2(\boldsymbol{AX}, \boldsymbol{AY}) &= \int_{\mathbb{R}^d} \mathbf{W}_2^2(\langle \boldsymbol{AX}, z/\|z\|\rangle, \langle \boldsymbol{AY}, z/\|z\|\rangle)\varphi(z)\mathrm{d}z \\ &= \int_{\mathbb{R}^d} \mathbf{W}_2^2(\langle \boldsymbol{X}, \boldsymbol{A}^\top z\rangle, \langle \boldsymbol{Y}, \boldsymbol{A}^\top z\rangle)\frac{\varphi(z)}{\|z\|^2}\mathrm{d}z.\end{aligned}$$

Via a change of variable $\tilde{\boldsymbol{z}} = \boldsymbol{A}^\top \boldsymbol{z}$ and after renaming and using the homogeneity property once more, we obtain

$$\begin{aligned}\mathbf{SW}_2^2(\boldsymbol{AX}, \boldsymbol{AY}) &= \int_{\mathbb{R}^d} \mathbf{W}_2^2(\langle \boldsymbol{X}, \tilde{z}\rangle, \langle \boldsymbol{Y}, \tilde{z}\rangle)\frac{\varphi(\boldsymbol{A}^{-\top}\tilde{z})}{\|\boldsymbol{A}^{-\top}\tilde{z}\|^2 |\det \boldsymbol{A}|}\mathrm{d}\tilde{z} \\ &= \int_{\mathbb{R}^d} \mathbf{W}_2^2(\langle \boldsymbol{X}, z/\|z\|\rangle, \langle \boldsymbol{Y}, z/\|z\|\rangle)\frac{\|z\|^2}{\|\boldsymbol{A}^{-\top}z\|^2}\frac{\varphi(\boldsymbol{A}^{-\top}z)}{|\det \boldsymbol{A}|}\mathrm{d}z.\end{aligned}$$

Hence, computing $\mathbf{SW}_2^2(\boldsymbol{AX}, \boldsymbol{AY})$, like computing $\mathbf{SW}_2^2(\boldsymbol{X}, \boldsymbol{Y})$, amounts to integrating

$$\mathbf{W}_2^2(\langle \boldsymbol{X}, z/\|z\|\rangle, \langle \boldsymbol{Y}, z/\|z\|\rangle)$$

over z, but with respect to the measure $\mu_{\boldsymbol{A}}$ that has the Lebesgue density

$$\mathrm{d}\mu_{\boldsymbol{A}}(z) = \frac{\|z\|^2}{\|\boldsymbol{A}^{-\top}z\|^2}\frac{\varphi(\boldsymbol{A}^{-\top}z)}{|\det \boldsymbol{A}|}\mathrm{d}z$$

instead of with respect to the measure μ that has the Lebesgue density

$$\mathrm{d}\mu(z) = \varphi(z)\mathrm{d}z.$$

Remark 5.2 (Radon–Nikodym derivative)**.** From the considerations above, we see that the Radon–Nikodym derivative of $\mu_{\boldsymbol{A}}$ with respect to μ is

$$\begin{aligned}\frac{\mathrm{d}\mu_{\boldsymbol{A}}}{\mathrm{d}\mu}(\boldsymbol{z}) &= \frac{\|\boldsymbol{z}\|^2}{\|\boldsymbol{A}^{-\top}\boldsymbol{z}\|^2}\frac{\varphi(\boldsymbol{A}^{-\top}\boldsymbol{z})}{|\det \boldsymbol{A}|\varphi(\boldsymbol{z})} \\ &= \frac{\|\boldsymbol{z}\|^2}{\boldsymbol{z}^\top(\boldsymbol{A}^\top\boldsymbol{A})^{-1}\boldsymbol{z}}\frac{1}{|\det \boldsymbol{A}|}\exp\left\{\frac{1}{2}\left(\boldsymbol{z}^\top\left(\boldsymbol{I}_d - (\boldsymbol{A}^\top\boldsymbol{A})^{-1}\right)\boldsymbol{z}\right)\right\}.\end{aligned}$$

♢

Remark 5.3 (Only $A^\top A$ matters)**.** We can write

$$|\det A| = \sqrt{|\det A||\det A|} = \sqrt{|\det A^\top||\det A|} = \sqrt{|\det A^\top A|}.$$

This shows that the Radon–Nikodym derivative obtained in the previous remark depends only on $A^\top A$, not on A itself. In particular, when whitening the input distributions using a matrix W with $W^\top W = \widehat{\Sigma}^{-1}$, the precise whitening procedure (e.g., Mahalanobis, PCA, Cholesky) does not affect the resulting sliced Wasserstein distances.[1]

More generally, consider a singular value decomposition $A = U_{\mathrm{SVD}}\Sigma_{\mathrm{SVD}}V_{\mathrm{SVD}}^\top$. Then $A^\top A$ has the eigenvalue decomposition $A^\top A = V_{\mathrm{SVD}}\Sigma_{\mathrm{SVD}}^2 V_{\mathrm{SVD}}^\top$, with V_{SVD} an orthogonal matrix and Σ_{SVD}^2 a diagonal matrix with the eigenvalues of $A^\top A$ on its main diagonal. The left singular vectors in U_{SVD}, then, are irrelevant for the sliced Wasserstein distance computation: for fixed $\Sigma_{\mathrm{SVD}}, V_{\mathrm{SVD}}$, any orthogonal U_{SVD} will result in the same distances. So we may assume that $A = \Sigma_{\mathrm{SVD}}V_{\mathrm{SVD}}^\top$, where Σ_{SVD} governs the degree of anisotropy of σ_A and $V_{\mathrm{SVD}}^\top$ the orientation. $\diamondsuit$

5.2 Geometrical interpretation

Consider the eigenvalue decomposition $A^\top A = Q\Lambda Q^\top$, where $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$. Then

$$z^\top (A^\top A)^{-1} z = (Q^\top z)^\top \Lambda^{-1} (Q^\top z) = \sum_{i=1}^d \lambda_i^{-1} (Q^\top z)_i^2.$$

Further note that

$$(\det A)^2 = \det(A^\top A) = \prod_{i=1}^d \lambda_i,$$

so

$$1/|\det A| = \prod_{i=1}^d \sqrt{\lambda_i^{-1}}.$$

The Radon–Nikodym derivative of μ_A with respect to μ can thus be written as

$$\frac{\mathrm{d}\mu_A}{\mathrm{d}\mu}(z) = \|z\|^2 \frac{\prod_{i=1}^d \sqrt{\lambda_i^{-1}}}{\sum_{i=1}^d \lambda_i^{-1}(Q^\top z)_i^2} \exp\left\{\frac{1}{2}\sum_{i=1}^d (1-\lambda_i^{-1})(Q^\top z)_i^2\right\}.$$

Write $\tilde{z} = Q^\top z$, that is, express z in terms of an orthonormal eigenbasis of $A^\top A$. Then

$$\frac{\mathrm{d}\mu_A}{\mathrm{d}\mu}(\tilde{z}) = \|\tilde{z}\|^2 \frac{\prod_{i=1}^d \sqrt{\lambda_i^{-1}}}{\sum_{i=1}^d \lambda_i^{-1}\tilde{z}_i^2} \exp\left\{\frac{1}{2}\sum_{i=1}^d (1-\lambda_i^{-1})\tilde{z}_i^2\right\}.$$

To appreciate the geometrical meaning of this, let us consider $\lambda_i = 1$ for $i = 2, \ldots, d$. Then, for $Q = (q_1, \ldots, q_d)$,

$$\frac{\mathrm{d}\mu_A}{\mathrm{d}\mu}(q_1) = \sqrt{\lambda_1} \exp\left\{\frac{1 - 1/\lambda_1}{2}\right\}$$

[1] But it *does* affect Wasserstein distances of the form $\mathbf{W}_2(\langle X, \theta\rangle, \langle Y, \theta\rangle)$ for $\theta \in \mathbb{R}^d$. As we will see in the real-data example at the end of this chapter, the choice of the whitening procedure can very much affect the performance of the downstream prediction model if Wasserstein kernels are used.

and

$$\frac{\mathrm{d}\mu_{\boldsymbol{A}}}{\mathrm{d}\mu}(\boldsymbol{q}_i) = \frac{1}{\sqrt{\lambda_1}},$$

$i = 2, \ldots, d$. Hence, for $\lambda_1 = 1$,

$$\frac{\mathrm{d}\mu_{\boldsymbol{A}}}{\mathrm{d}\mu}(\boldsymbol{q}_1) = \frac{\mathrm{d}\mu_{\boldsymbol{A}}}{\mathrm{d}\mu}(\boldsymbol{q}_i) = 1;$$

for $\lambda_1 > 1$,

$$\frac{\mathrm{d}\mu_{\boldsymbol{A}}}{\mathrm{d}\mu}(\boldsymbol{q}_1) > \sqrt{\lambda_1} > 1 \text{ and } \frac{\mathrm{d}\mu_{\boldsymbol{A}}}{\mathrm{d}\mu}(\boldsymbol{q}_i) < 1;$$

and for $\lambda_1 < 1$,

$$\frac{\mathrm{d}\mu_{\boldsymbol{A}}}{\mathrm{d}\mu}(\boldsymbol{q}_1) < \sqrt{\lambda_1} < 1 \text{ and } \frac{\mathrm{d}\mu_{\boldsymbol{A}}}{\mathrm{d}\mu}(\boldsymbol{q}_i) > 1.$$

So, relative to the isotropic measure μ, the measure $\mu_{\boldsymbol{A}}$ assigns more weight to the directions aligned with eigenvectors $\boldsymbol{q}_i$ for which $\lambda_i > 1$ and less weight to the directions aligned with eigenvectors $\boldsymbol{q}_i$ for which $\lambda_i < 1$.

Remark 5.4. For $\boldsymbol{A} \in \mathrm{GL}_d(\mathbb{R})$, the set

$$\mathcal{E}_{\boldsymbol{A}} := \{\boldsymbol{A}^\top \boldsymbol{\theta} : \boldsymbol{\theta} \in \mathbb{S}^{d-1}\}$$

is the surface of an ellipsoid in $\mathbb{R}^d$. However, while σ is the uniform distribution on $\mathbb{S}^{d-1}$, the pushforward measure $\sigma_{\boldsymbol{A}}$ is not the uniform distribution on $\mathcal{E}_{\boldsymbol{A}}$. This is illustrated for the two-dimensional case in Figure 5.1, for $\boldsymbol{A} = \begin{pmatrix} 2 & 1 \\ 1/2 & 1 \end{pmatrix}$. The left panel shows 36 equally spaced points on the unit sphere, the arc segment between each neighbouring pair having probability $1/36$ under σ. The map $\boldsymbol{\theta} \mapsto \boldsymbol{A}^\top\boldsymbol{\theta}$ sends these points to $\mathcal{E}_{\boldsymbol{A}}$, with the arc segments between consecutive points on the ellipse also having probability $1/36$ under $\sigma_{\boldsymbol{A}}$. Clearly, the image points are not equally spaced along the ellipse: they cluster more tightly near directions aligned with the eigenvectors of $\boldsymbol{A}^\top\boldsymbol{A}$ that have eigenvalues greater than 1 (i.e., along the elongated elliptical axis) and are more spread out near directions aligned with eigenvectors with small eigenvalues (i.e., along the compressed elliptical axis).

For comparison, the right panel shows what the uniform distribution on $\mathcal{E}_{\boldsymbol{A}}$ looks like. The 36 points on the ellipse are equally spaced along the ellipse, with each arc segment between neighbouring points having probability $1/36$ under $\mathrm{Uniform}(\mathcal{E}_{\boldsymbol{A}})$. Mapping these points to the unit sphere via the map $\boldsymbol{\theta} \mapsto \boldsymbol{A}^{-\top}\boldsymbol{\theta}$, we can see that resulting distribution concentrates more probability mass near directions aligned with eigenvectors with small eigenvalues than does $\sigma_{\boldsymbol{A}}$.

Spelling out precisely how $\sigma_{\boldsymbol{A}}$ relates to the uniform distribution on $\mathcal{E}_{\boldsymbol{A}}$ by means of a Radon–Nikodym derivative would require computing surface integrals of $\mathcal{E}_{\boldsymbol{A}}$. This is non-trivial even for $d = 2, 3$. I refer the interested reader to Marples & Williams (2024) for some results that may prove useful for the three-dimensional case. ♢

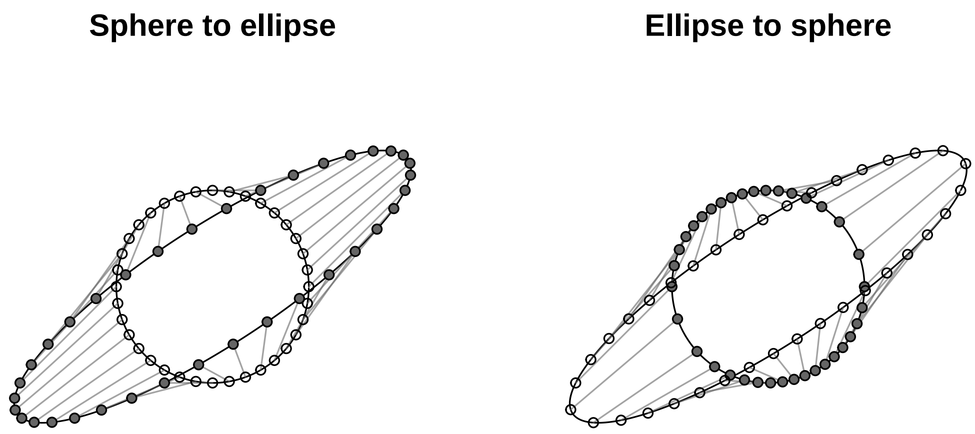

Figure 5.1: The pushforward measure $\sigma_{\boldsymbol{A}}$ is not the uniform distribution on an ellipse. *Left:* The uniform distribution on the sphere pushed forward to the ellipse using the map $\boldsymbol{\theta} \mapsto \boldsymbol{A}^\top \boldsymbol{\theta}$. *Right:* The uniform distribution on the ellipse pushed forward to the sphere using the map $\boldsymbol{\theta} \mapsto \boldsymbol{A}^{-\top} \boldsymbol{\theta}$. In this example, $\boldsymbol{A} = \begin{pmatrix} 2 & 1 \\ 1/2 & 1 \end{pmatrix}$.

5.3 Computational consequences

Equation 5.1 on page 36 reveals two ways in which the computation of sliced Wasserstein distances between distributions subjected to a global affine transformation may be simplified.

Remark 5.5 (Translations are unnecessary)**.** If researchers want to scale, whiten or otherwise reshape the input distributions by means of a global bijective affine transformation, Equation 5.1 shows that it is not necessary to carry out the translation step: The sliced 2-Wasserstein distance between the distributions of $\boldsymbol{A}(\boldsymbol{X} - \boldsymbol{v})$ and $\boldsymbol{A}(\boldsymbol{Y} - \boldsymbol{v})$ is the same as the one between the distributions of $\boldsymbol{AX}$ and $\boldsymbol{AY}$, for any matrix $\boldsymbol{A}$ of full rank and any vector $\boldsymbol{v} \in \mathbb{R}^d$. This shaves off an admittedly cheap step from each transformation. $\diamondsuit$

Remark 5.6 (Time complexity of Monte Carlo simulations)**.** Let $\boldsymbol{M}_1, \ldots, \boldsymbol{M}_N \in \mathbb{R}^{n \times d}$ be matrices representing the empirical input distributions. We wish to use Monte Carlo simulation to estimate the pairwise sliced Wasserstein distances between these distributions after subjecting them to a global affine transformation. To this end, we may compute

$$\widehat{\mathbf{SW}}_2^2(\boldsymbol{M}_i\boldsymbol{A}, \boldsymbol{M}_j\boldsymbol{A}; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L) = \frac{1}{L} \sum_{\ell=1}^{L} \mathbf{W}_2^2(\langle \boldsymbol{M}_i\boldsymbol{A}, \boldsymbol{\theta}_\ell \rangle, \langle \boldsymbol{M}_j\boldsymbol{A}, \boldsymbol{\theta}_\ell \rangle)$$

with stochastically independent $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_L \sim \mathrm{Uniform}(\mathbb{S}^{d-1})$ for some integer L. This is equivalent to computing

$$\frac{1}{L} \sum_{\ell=1}^{L} \mathbf{W}_2^2(\langle \boldsymbol{M}_i, \boldsymbol{A}^\top \boldsymbol{\theta}_\ell \rangle, \langle \boldsymbol{M}_j, \boldsymbol{A}^\top \boldsymbol{\theta}_\ell \rangle).$$

The time complexity of computing $\langle \boldsymbol{M}_i\boldsymbol{A}, \boldsymbol{\theta}_\ell\rangle$ for $i = 1, \dots, N$, $\ell = 1, \dots, L$, that is, of first transforming the matrices and then computing the projections, is in $\mathcal{O}(Nnd^2 + NnLd)$: Each of the N matrix multiplications is in $\mathcal{O}(nd^2)$, and each of the N times L matrix-by-vector multiplications is in $\mathcal{O}(nd)$. By contrast, the time complexity of computing $\langle \boldsymbol{M}_i, \boldsymbol{A}^\top\boldsymbol{\theta}_\ell\rangle$, $i = 1, \dots, N$, $\ell = 1, \dots, L$, that is, of first transforming the projection directions rather than the matrices, is in $\mathcal{O}(Ld^2 + NnLd)$: For each projection direction, computing $\boldsymbol{A}^\top\boldsymbol{\theta}_\ell$ is in $\mathcal{O}(d^2)$, and each of the N times L matrix-by-vector multiplications is still in $\mathcal{O}(nd)$.

Hence, if L is small relative to Nn, applying the transformation to the projection directions instead of to the data matrices may speed up computations somewhat. In typical applications $L \ll Nn$, so transforming the projection directions rather than the input distributions is almost always preferable. That said, after implementing this suggestion, the overall time complexity for computing $\widehat{\mathbf{SW}}_2^2(\boldsymbol{M}_i\boldsymbol{A}, \boldsymbol{M}_j\boldsymbol{A}; \boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_L)$ for all $i < j$ is in

$$\mathcal{O}(Ld^2 + LdNn\log n + LN^2n),$$

where the second term is due to computing and then sorting $\langle \boldsymbol{M}_i, \boldsymbol{A}^\top\boldsymbol{\theta}_\ell\rangle \in \mathbb{R}^n$ for $i = 1, \dots, N$, $\ell = 1, \dots, L$, and the third term is due to computing the sum of the squared element-wise differences between these sorted vectors for $1 \leq i < j \leq N$ and $\ell = 1, \dots, L$. In practice, the first term will often be dwarfed by the other two. ♢

Remark 5.7 (On reweighting directions)**.** Given independent $\boldsymbol{Z}_1, \dots, \boldsymbol{Z}_L \sim \mathcal{N}_d(\boldsymbol{0}, \boldsymbol{I}_d)$, the squared sliced Wasserstein distance $\mathbf{SW}_2^2(\boldsymbol{M}_i, \boldsymbol{M}_j)$ may be estimated as

$$\frac{1}{L}\sum_{\ell=1}^{L} \mathbf{W}_2^2(\langle \boldsymbol{M}_i, \boldsymbol{Z}_\ell/\|\boldsymbol{Z}_\ell\|\rangle, \langle \boldsymbol{M}_j, \boldsymbol{Z}_\ell/\|\boldsymbol{Z}_\ell\|\rangle).$$

If the realisations of $(\boldsymbol{Z}_\ell)_{\ell=1}^L$ as well as the distances $(\mathbf{W}_2^2(\langle \boldsymbol{M}_i, \boldsymbol{Z}_\ell/\|\boldsymbol{Z}_\ell\|\rangle, \langle \boldsymbol{M}_j, \boldsymbol{Z}_\ell/\|\boldsymbol{Z}_\ell\|\rangle))_{\ell=1}^L$ are kept available, the theoretical considerations suggest that they may be reused to more quickly estimate $\mathbf{SW}_2^2(\boldsymbol{M}_i\boldsymbol{A}, \boldsymbol{M}_j\boldsymbol{A})$ for any $\boldsymbol{A} \in \mathrm{GL}_d(\mathbb{R})$ by means of importance sampling using the Radon–Nikodym derivative:

$$\frac{1}{L}\sum_{\ell=1}^{L} \mathbf{W}_2^2(\langle \boldsymbol{M}_i, \boldsymbol{Z}_\ell/\|\boldsymbol{Z}_\ell|\rangle, \langle \boldsymbol{M}_j, \boldsymbol{Z}_\ell/\|\boldsymbol{Z}_\ell\|\rangle)\frac{\mathrm{d}\mu_{\boldsymbol{A}}}{\mathrm{d}\mu}(\boldsymbol{Z}_\ell).$$

However, preliminary attempts failed due to numerical issues and an often dramatic reduction of the effective sample size. I therefore leave a more systematic exploration of this topic as an avenue for future work. ♢

5.4 Consequences for kernel learning

We have seen that transforming the input distributions by a global affine bijection corresponds to a change of measure in the sliced Wasserstein computations. We now consider the effects of this action on the downstream task of constructing a kernel from the resulting sliced Wasserstein distances. To this end, assume that we have estimated

$$\mathbf{SW}_2^2(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j) = \int \mathbf{W}_2^2(\langle \boldsymbol{X}, \boldsymbol{A}^\top \theta \rangle, \langle \boldsymbol{Y}, \boldsymbol{A}^\top \theta \rangle) \mathrm{d}\sigma(\theta),$$

for random vectors $\boldsymbol{X}_i \sim P_i \in \mathcal{P}_2(\mathbb{R}^d), i = 1, \ldots, N$, and $\boldsymbol{A} \in \mathrm{GL}_d(\mathbb{R})$.

When constructing the sliced Wasserstein kernel $k_{SW}(\cdot, \cdot)$ with scaling factor $s^2 > 0$ and length-scale $t > 0$, we observe that

$$\begin{aligned} k_{SW}(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j) &= s^2 \exp\left\{ -\frac{1}{2t^2} \mathbf{SW}_2^2(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j) \right\} \\ &= s^2 \exp\left\{ -\frac{1}{2} \mathbf{SW}_2^2 \left(\frac{1}{t}\boldsymbol{A}\boldsymbol{X}_i, \frac{1}{t}\boldsymbol{A}\boldsymbol{X}_j \right) \right\}. \end{aligned}$$

Any choice of a positive $t \neq 1$ thus corresponds to multiplying the input distributions by a common factor. Equivalently, it corresponds to integrating $\mathbf{W}_2^2(\langle \boldsymbol{X}_i, \boldsymbol{A}^\top \theta \rangle, \langle \boldsymbol{X}_j, \boldsymbol{A}^\top \theta \rangle)$ over θ with respect to the measure Uniform$(t^{-1}\mathbb{S}^{d-1})$ or to using $t^{-1}\boldsymbol{A}$ as the transformation matrix instead of $\boldsymbol{A}$.

In applications, the length-scale t is typically a hyperparameter tuned using training data. If the transformation matrix $\boldsymbol{A} = \boldsymbol{\Sigma}_{\mathrm{SVD}} \boldsymbol{V}_{\mathrm{SVD}}^\top$ is learnt alongside t, we cannot uniquely identify $\boldsymbol{A}$ and t as only the product $t^{-1}\boldsymbol{A}$, or more accurately: the product $t^{-2}\boldsymbol{A}^\top \boldsymbol{A}$, matters. This suggests that it would be sensible to normalise the transformation matrix $\boldsymbol{A}$ in such a way that $\varsigma(\boldsymbol{A}) = 1$, for some scale $\varsigma : \mathrm{GL}_d(\mathbb{R}) \to (0, \infty)$. The Radon–Nikodym derivative of $\mu_{\boldsymbol{A}}$ with respect to μ obtained earlier suggests the scale map

$$\varsigma(\boldsymbol{A}) = |\det \boldsymbol{A}| = \prod_{i=1}^d \sqrt{\lambda_i}.$$

For this scale map, we may normalise $\tilde{\boldsymbol{A}} \in \mathrm{GL}_d(\mathbb{R})$ by computing $\boldsymbol{A} = |\det(\tilde{\boldsymbol{A}})|^{-1/d}\tilde{\boldsymbol{A}}$. That said, other maps may be useful, too (e.g., matrix norms). With such a restriction in place, learning a suitable $\boldsymbol{A}$ amounts to learning the appropriate anisotropy and orientation of $\mu_{\boldsymbol{A}}$ relative to μ, whereas tuning t amounts to learning an appropriate scale.

5.5 Behaviour under extreme anisotropy

It seems intuitively appealing to use global affine transformations to upweight those directions in the input distributions that are believed to be most informative for the downstream task of predicting the outcome variable, while downweighting those that carry primarily noise. As the present section will show, however, transforming the input data by an extremely anisotropic affine transformation (even one aligned with the signal) carries a risk of shooting oneself in the foot, as this can lead to degenerate sliced Wasserstein distances and, as a result, degenerate kernel matrices.

5.5.1 Asymptotic considerations

Let us investigate the effects of the anisotropy of the transformation matrix $\boldsymbol{A}$ on sliced Wasserstein distances and the resulting kernels. For simplicity, we ignore any Monte Carlo variance, that is, we consider the distances $\mathbf{SW}_2^2(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j)$ and the kernel values $k_{ij} := k_{SW}(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j)$. A further simplifying assumption is that $\boldsymbol{X}_i \sim P_i \in \mathcal{P}_2(\mathbb{R}^2)$ for $i = 1, \dots, N$. Moreover, we consider a diagonal transformation matrix $\boldsymbol{A}$: As our earlier results show, we may assume that $\boldsymbol{A} = \boldsymbol{\Sigma}_{\text{SVD}}\boldsymbol{V}_{\text{SVD}}^\top$, with $\boldsymbol{\Sigma}_{\text{SVD}}$ diagonal and $\boldsymbol{V}_{\text{SVD}}$ an orthogonal matrix. By focusing on a diagonal matrix $\boldsymbol{A}$, we isolate the degree of anisotropy of the transformation encoded by $\boldsymbol{\Sigma}_{\text{SVD}}$ from its orientation as encoded by $\boldsymbol{V}_{\text{SVD}}$.

We will consider two scenarios. In the first, the transformation matrices considered have the form $\boldsymbol{A} = \text{diag}(\alpha, 1)$ for some $\alpha > 0$. In the second, we consider transformation matrices $\boldsymbol{A}$ such that $\varsigma(\boldsymbol{A}) = |\det \boldsymbol{A}| = 1$, namely, transformation matrices of the type $\boldsymbol{A} = \text{diag}(\sqrt{\alpha}, 1/\sqrt{\alpha})$ for some $\alpha > 0$.

Unnormalised transformation matrices. We first consider an unnormalised transformation matrix $\boldsymbol{A} = \text{diag}(\alpha, 1)$ for $\alpha > 0$. It holds that

$$\begin{aligned}
\mathbf{SW}_2^2(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j) &= \int \mathbf{W}_2^2(\langle \boldsymbol{A}\boldsymbol{X}_i, \theta\rangle, \langle \boldsymbol{A}\boldsymbol{X}_j, \theta\rangle)\mathrm{d}\sigma(\theta) \\
&= \int \mathbf{W}_2^2(\alpha\theta_1 X_{i1} + \theta_2 X_{i2}, \alpha\theta_1 X_{j1} + \theta_2 X_{j2})\mathrm{d}\sigma(\theta) \\
&= \alpha^2 \int \theta_1^2 \mathbf{W}_2^2\left(X_{i1} + \frac{\theta_2}{\alpha\theta_1}X_{i2}, X_{j1} + \frac{\theta_2}{\alpha\theta_1}X_{j2}\right)\mathrm{d}\sigma(\theta).
\end{aligned}$$

Since $\theta_1^2, \theta_2^2 \in (0,1)$ σ-almost everywhere, it follows from Property 3 presented in Lemma 2.2 on page 8 and the inequality $(a+b)^2 \leq 2a^2 + 2b^2$ that, for all $\alpha \geq 1$,

$$\begin{aligned}
\theta_1^2 \mathbf{W}_2^2\left(X_{i1} + \frac{\theta_2}{\alpha\theta_1}X_{i2}, X_{j1} + \frac{\theta_2}{\alpha\theta_1}X_{j2}\right) &\leq 2\theta_1^2\left(\mathbb{E}((X_{i1} + \frac{\theta_2}{\alpha\theta_1}X_{i2})^2) + \mathbb{E}((X_{j1} + \frac{\theta_2}{\alpha\theta_1}X_{j2})^2)\right) \\
&\leq 4\theta_1^2\left(\mathbb{E}(X_{i1}^2 + X_{j1}^2) + \left(\frac{\theta_2}{\alpha\theta_1}\right)^2 \mathbb{E}(X_{i2}^2 + X_{j2}^2)\right) \\
&\leq 4\left(\mathbb{E}(X_{i1}^2 + X_{j1}^2) + \frac{1}{\alpha^2}\mathbb{E}(X_{i2}^2 + X_{j2}^2)\right) \\
&\leq 4\left(\mathbb{E}(X_{i1}^2 + X_{j1}^2) + \mathbb{E}(X_{i2}^2 + X_{j2}^2)\right) \\
&= 4\left(\mathbb{E}(\|\boldsymbol{X}_i\|^2) + \mathbb{E}(\|\boldsymbol{X}_j\|^2)\right) \\
&=: C,
\end{aligned}$$

σ-almost everywhere, with $\int C\mathrm{d}\sigma(\theta) = C < \infty$. Further, due to the continuity of the Wasserstein distance (see Lemma 2.3 on page 9),

$$\mathbf{W}_2\left(X_{i1} + \frac{\theta_2}{\alpha\theta_1}X_{i2}, X_{j1} + \frac{\theta_2}{\alpha\theta_1}X_{j2}\right) \to \mathbf{W}_2\left(X_{i1}, X_{j1}\right)$$

as $\alpha \to \infty$. By the dominated convergence theorem, then, it follows that

$$\begin{aligned}
\lim_{\alpha\to\infty} \mathbf{SW}_2^2(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j) &= \left(\lim_{\alpha\to\infty}\alpha^2\right)\mathbf{W}_2^2(X_{i1}, X_{j1})\int \theta_1^2\mathrm{d}\sigma(\theta) \\
&= \left(\lim_{\alpha\to\infty}\alpha^2\right)\frac{\mathbf{W}_2^2(X_{i1}, X_{j1})}{2}.
\end{aligned}$$

In the last step, I used a symmetry argument:

$$2\int \theta_1^2 \mathrm{d}\sigma(\theta) = \int \theta_1^2 \mathrm{d}\sigma(\theta) + \int \theta_2^2 \mathrm{d}\sigma(\theta) = \int \theta_1^2 + \theta_2^2 \mathrm{d}\sigma(\theta) = \int 1 \mathrm{d}\sigma(\theta) = 1.$$

Hence, as $\alpha \to \infty$,

$$\mathbf{SW}_2^2(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j) \to \begin{cases} 0, & \text{if } \mathcal{L}(X_{i1}) = \mathcal{L}(X_{j1}), \\ \infty, & \text{otherwise.} \end{cases}$$

By the continuity of the kernel function, we thus find that

$$k_{ij} \to s^2[\mathcal{L}(X_{i1}) = \mathcal{L}(X_{j1})]$$

as $\alpha \to \infty$.

Assuming that $\mathcal{L}(\boldsymbol{X}_i) \neq \mathcal{L}(\boldsymbol{X}_j)$ whenever $i \neq j$, then, the sliced Wasserstein kernel matrix degenerates towards $s^2\boldsymbol{I}_N$ as $\alpha \to \infty$. At asymptote, the Gaussian process model will therefore assign the mean of the training outcomes to the test distributions without picking up on any signal. However, increasing the length-scale t in lockstep with α would offset this degeneracy in practice.

Let us now consider what happens when $\alpha \downarrow 0$. Using the same techniques as above, we find that, for all $\alpha \in (0, 1]$,

$$\begin{aligned} \mathbf{W}_2^2(\alpha\theta_1 X_{i1} + \theta_2 X_{i2}, \alpha\theta_1 X_{j1} + \theta_2 X_{j2}) &\leq 4\left(\alpha^2\theta_1^2 \mathbb{E}(X_{i1}^2 + X_{j1}^2) + \theta_2^2 \mathbb{E}(X_{i2}^2 + X_{j2}^2)\right) \\ &\leq 4\mathbb{E}\left(X_{i1}^2 + X_{j1}^2 + X_{i2}^2 + X_{j2}^2\right) \\ &= C, \end{aligned}$$

for all $\alpha \in (0, 1]$ σ-almost everywhere. Hence, we may apply the dominated convergence theorem to find that

$$\begin{aligned} \lim_{\alpha\downarrow 0} \mathbf{SW}_2^2(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j) &= \lim_{\alpha\downarrow 0} \int \mathbf{W}_2^2(\alpha\theta_1 X_{i1} + \theta_2 X_{i2}, \alpha\theta_1 X_{j1} + \theta_2 X_{j2}) \mathrm{d}\sigma(\theta) \\ &= \int \mathbf{W}_2^2(\theta_2 X_{i2}, \theta_2 X_{j2}) \mathrm{d}\sigma(\theta) \\ &= \frac{\mathbf{W}_2^2(X_{i2}, X_{j2})}{2}. \end{aligned}$$

Thus, at asymptote, the resulting sliced Wasserstein kernel matrix will not be degenerate as $\alpha \downarrow 0$. It will, however, be sensitive only to distributional differences along the second dimension.

Normalised transformation matrices. We retain the setting from earlier but use a normalised transformation matrix $\boldsymbol{A} = \mathrm{diag}(\sqrt{\alpha}, 1/\sqrt{\alpha})$. Analogous calculations show that

$$\lim_{\alpha\to\infty} \mathbf{SW}_2^2(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j) = \left(\lim_{\alpha\to\infty} \alpha\right) \frac{\mathbf{W}_2^2(X_{i1}, X_{j1})}{2} = \begin{cases} 0, & \text{if } \mathcal{L}(X_{i1}) = \mathcal{L}(X_{j1}), \\ \infty, & \text{otherwise.} \end{cases}$$

In contrast to when using an unnormalised transformation matrix, however, the rate of convergence to degeneracy is linear in α rather than quadratic. Hence, choosing t proportionally to $\sqrt{\alpha}$ would offset the degeneracy in practice.

Similar calculations reveal that

$$\lim_{\alpha\downarrow 0} \mathbf{SW}_2^2(\boldsymbol{A}\boldsymbol{X}_i, \boldsymbol{A}\boldsymbol{X}_j) = \lim_{\alpha\downarrow 0}\left(\frac{1}{\alpha}\right)\frac{\mathbf{W}_2^2(X_{i2}, X_{j2})}{2} = \begin{cases} 0, & \text{if } \mathcal{L}(X_{i2}) = \mathcal{L}(X_{j2}), \\ \infty, & \text{otherwise.} \end{cases}$$

In this case, choosing t proportionally to $\alpha^{-1/2}$ would offset the degeneracy in practice.

5.5.2 Illustrations

In the remainder of this section, I simulate and analyse data under two scenarios. In the first, only the first dimension contains a signal of the outcome variable; in the second, the first dimension contains most of the signal of the outcome variable, but the second dimension contains some signal, too.

Set-up. We consider $(P_i, y_i)_{i=1}^{100}$, where

$$P_i = \mathcal{N}_2(\boldsymbol{\mu}_i, \boldsymbol{I}_2)$$

and

$$y_i = f(\mu_{i1}, \mu_{i2}),$$

for a function $f : \mathbb{R}^2 \to \mathbb{R}$ that will be introduced shortly, $i = 1, \dots, 100$. From each distribution P_i, a simple random sample with 200 observations is drawn. These observations are represented in a matrix $\boldsymbol{M}_i \in \mathbb{R}^{200\times 2}$. Given $(\boldsymbol{M}_i, y_i)_{i=1}^{50}$ and $(\boldsymbol{M}_i)_{i=51}^{100}$, the aim is to predict $y_{51}, \dots, y_{100}$.

The mean vectors $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_{100}$ are drawn independently from Uniform$((-\pi, \pi)^2)$. As for the generation of the outcomes $y_1, \dots, y_{100}$, two scenarios are considered. To this end, we define a helper function

$$\begin{aligned} g &: \mathbb{R} \to \mathbb{R}, \\ x &\mapsto \begin{cases} \frac{\cos(2\pi x) - \cos(\pi x)}{\pi x}, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0; \end{cases} \end{aligned}$$

see Figure 5.2. In the first scenario, the outcome data are generated as

$$y_i = f(\boldsymbol{\mu}_i) := 1.5 g(\mu_{i1});$$

in the second, they are generated as

$$y_i = f(\boldsymbol{\mu}_i) := g(\mu_{i1}) + \frac{1}{2} g(\mu_{i2}),$$

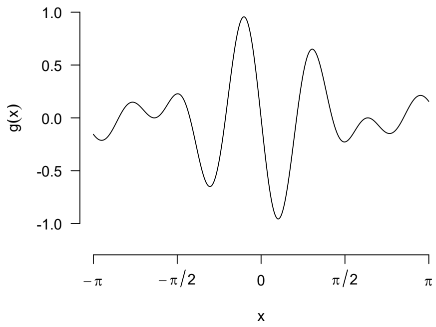

Figure 5.2: The function g used to generate the outcome data.

$i = 1, \ldots, 100$. That is, in the first scenario, the entire signal is contained in the distributions' first dimension; in the second, two-thirds of the signal is contained in the first dimension, and the remaining third is contained in the second dimension.

For both scenarios, the mean vectors and the empirical distributions are generated ten times. Each of the resulting ten data sets per scenario is analysed independently of the others.

Analysis. First, a baseline is computed by using the mean outcome in the training sample as the prediction for all outcomes:

$$\widehat{y}_i = \frac{1}{50} \sum_{j=1}^{50} y_j,$$

$i = 51, \ldots, 100$.

Second, Gaussian process models using sliced Wasserstein kernels were used. Specifically, the kernel matrix was constructed as

$$\boldsymbol{K} = (\widehat{k}_{SW}(\boldsymbol{M}_i, \boldsymbol{M}_j; \boldsymbol{A}^\top \boldsymbol{\theta}_1, \ldots, \boldsymbol{A}^\top \boldsymbol{\theta}_{50}))_{1 \le i,j \le 100},$$

for independent $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_{50} \sim \text{Uniform}(\mathbb{S}^1)$; see Remark 5.6 on page 40. The transformation matrix is constructed as

$$\boldsymbol{A} = c(\alpha) \begin{pmatrix} \alpha & 0 \\ 0 & 1 \end{pmatrix}$$

for $\alpha = 10^{-6}, \ldots, 10^{6}$. Both unnormalised transformation matrices (with $c(\alpha) \equiv 1$) and normalised transformation matrices (with $c(\alpha) = \alpha^{-1/2}$) are considered. Further, the Gaussian process models are fitted using both hyperparameter tuning as well as with fixed hyperparameters $\eta = 10^{-6}, t = 1, s^2 = 1$.

Finally, for each pair of matrices, I computed the kernel value

$$k(\boldsymbol{M}_i, \boldsymbol{M}_j; \boldsymbol{e}_1, \boldsymbol{e}_2),$$

see Definition 3.12 on page 23. This kernel combines the information from both marginals into a single kernel, potentially allowing the model to use both projection directions simultaneously.

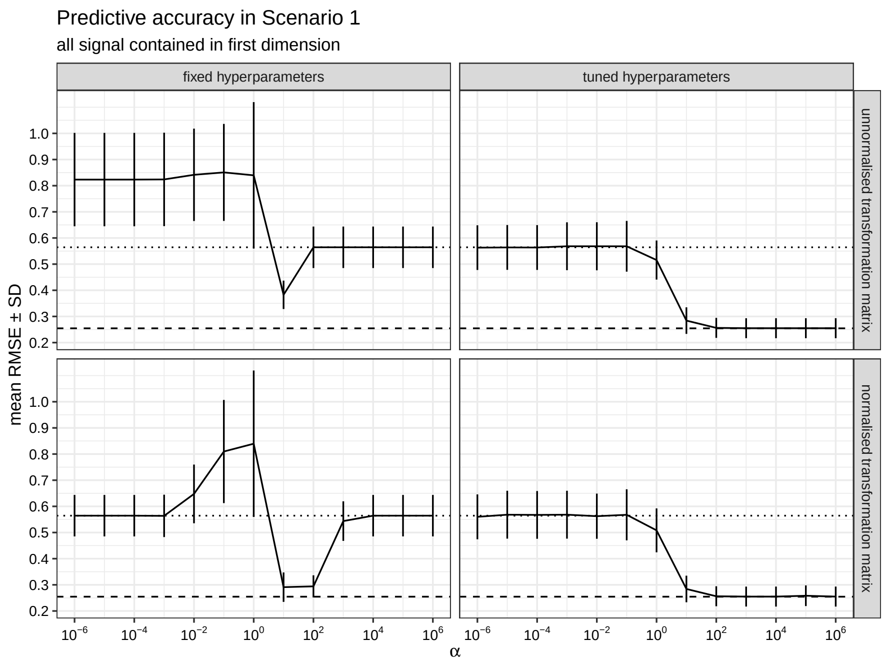

Figure 5.3: Mean and standard deviations of the root mean squared error obtained in the first scenario in ten runs. The dotted horizontal line around 0.55 shows the mean performance of a baseline model (training mean imputation); the dashed horizontal line around 0.25 shows the mean performance of a Gaussian process model based on Wasserstein distances along the cardinal axes only.

The kernel was used in another Gaussian process model, whose hyperparameters s_1^2, s_2^2, t_1, t_2 and η^2 were tuned as explained in Remark 3.13 on page 25.

Results. The results for the first scenario are shown in Figure 5.3. The top left panel shows that, if the transformation matrix is not normalised and the hyperparameters are not tuned, the model performs at baseline for large α and worse than baseline for small α. This is consistent with our theoretical considerations: For large α, the kernel matrix degenerates to a multiple of the identity, which would cause the Gaussian process model to assign the training sample mean to the test observations, leading to at-baseline performance. For small α, the kernel matrix does not degenerate but is dominated by noise instead.

The bottom left panel shows at-baseline performance for both small and large α when using a normalised transformation matrix without hyperparameter tuning. This, too, is in keeping with our finding that the kernel matrix degenerates to a multiple of the identity in both cases.

The panels on the right show that hyperparameter tuning offsets the kernel degeneracy for large α, regardless of the way the transformation matrix was constructed. For small α, hyperparameter tuning seems to be useful in the sense that it prevents worse-than-baseline performance. It is also worth noting that the sliced Wasserstein kernels do not outperform a kernel based on the Wasserstein distances along the cardinal axes.

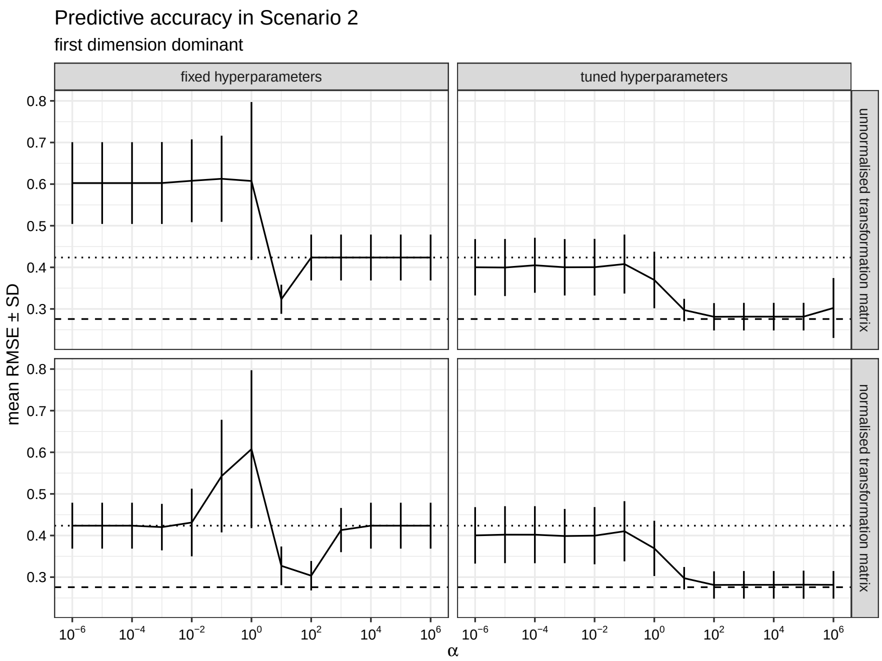

Figure 5.4: Mean and standard deviations of the root mean squared error obtained in the second scenario in ten runs. The dotted horizontal line around 0.42 shows the mean performance of a baseline model (training mean imputation); the dashed horizontal line around 0.27 shows the mean performance of a Gaussian process model based on Wasserstein distances along the cardinal axes only.

Let us now turn to the results for the second scenario, shown in Figure 5.4. When the hyperparameters are not tuned (panels on the left-hand side), the results are qualitatively similar to those in the first scenario. When the hyperparameters are tuned, however, the model is able to leverage the signal in the second dimension when α is small to beat the performance of the baseline model. This stands in contrast to the first scenario, where no above-baseline performance was achievable at small α even with tuning, since the second dimension carried no signal there. Upweighting the first dimension by using larger α, however, improves performance, but there is again little difference between using a moderately large α and an extremely large one. Again, the sliced Wasserstein kernels do not outperform a kernel based on the Wasserstein distances along the cardinal axes.

Conclusions. The simulations underscore the insights gleaned from the asymptotic analysis. Without hyperparameter tuning, extreme values of α negatively affect predictive accuracy. Hyperparameter tuning offsets the degeneracy found for large α values, allowing the model to beat the baseline. For small α, hyperparameter tuning prevents worse-than-baseline performance and may even help the model to beat it (in the second scenario). Of note, the use of normalised and unnormalised matrices yielded essentially identical results when the hyperparameters were tuned, suggesting that such tuning is able to offset scale differences between both kinds of transformation matrix.

However, the performance achieved by the models using sliced Wasserstein kernels did not improve upon that of a kernel constructed using only the Wasserstein distances along the two cardinal axes. This finding contrasts with the one obtained for the toy example in Chapter 4. The difference between both simulations is that in the present chapter, the signal is entirely axis-aligned, whereas this was not the case in the previous chapter. A sum-of-kernels regression model may therefore serve as a useful point of comparison when gauging the extent to which the prediction is improved by considering the full distribution.

5.6 A real-data example

The explorations in the present chapter were spurred by my attempt to reanalyse a dataset collected by Eckermann et al. (2021) using sliced Wasserstein distances. During this attempt, I wondered if and how the input data should be scaled or whitened. In this section, I will use Eckermann et al.'s dataset in order to illustrate the effects of scaling and whitening on a prediction task using sliced Wasserstein distances. Moreover, this dataset serves as a further, more realistic test case for gauging the added value of sliced Wasserstein distances over merely using Wasserstein distances along the cardinal axes.

5.6.1 Data and objective

Eckermann et al. (2021) studied the hippocampi of $N = 20$ deceased individuals with a view to identifying architectural differences between the hippocampi of individuals with and without Alzheimer's disease. Their data extraction method is too complex to summarise here, but in brief, each of the twenty hippocampi was represented by an $n_i \times 5$ matrix, with the number of rows varying from 2,748 to 37,348 (median: 17,069). Each row represents a neuron through the following five measurements:

- its nucleus compactness, as operationalised by the median of the electron density in the neuron. Compactness is not used in its mathematical sense here;
- the normalised variance of the electron density within the nucleus;
- the neuron's volume in cubic micrometres;
- a measure of the nucleus's sphericity;
- the number of neighbours, i.e., of other neurons within 13.5 μm of the neuron.

While Eckermann et al. (2021) were interested in differences between individuals with and without Alzheimer's disease, the goal that I will pursue is to assess to what extent the distributional information in the first four of these measurements can be used to retrieve two scalar summary statistics based on the fifth one, namely (1) the 10% trimmed mean of the number of neighbours for each hippocampus and (2) the entropy of the distribution of the number of neighbours for each hippocampus. Denoting by $\widehat{p}_k$ the proportion of neurons in a given hippocampus for which the number of neighbours equals k, this entropy is computed as $-\sum_{k\geq 0} \widehat{p}_k \log \widehat{p}_k$, with the convention that $0 \log 0 = 0$.

5.6.2 Analysis

Data preparation. I first applied the same transformation and filtering steps as Eckermann et al. (2021), resulting in a dataset comprising $\sum_{i=1}^{20} n_i = 344{,}420$ neurons. Ten of the neurons had negative compactness values. I removed these on the assumption that these must have been caused by some transcription error, but their impact on the results below is presumably negligible. Noticing that the volume variable spanned several orders of magnitude across the matrices (from 35 to 19,958), I $\log_{10}$-transformed this variable.

For the prediction task, the input matrix $\boldsymbol{M}_i \in \mathbb{R}^{n_i \times 4}$ containing the n_i compactness, density variance, volume, and sphericity measurements for the i-th hippocampus's neurons serve as the input and are interpreted as representing an empirical distribution $\widehat{P}_i$. Two scalar summaries of the distributions of the number of neighbours will serve as outcomes: the 10% trimmed mean of the number of neighbours observed for the i-th hippocampus, and the entropy of the distribution of the number of neighbours observed for the i-th hippocampus.

Model assessment. Since the number of input matrices is small ($N = 20$), I used leave-one-out crossvalidation (LOOCV) to assess the predictive utility of the modelling techniques used. More precisely, the outcome for the i-th hippocampus was predicted by a model for which only $(\boldsymbol{M}_j, y_j)_{j \neq i}$ were used as training data. This also means that in order to generate a prediction for the i-th outcome, the scaling or whitening matrices (see below) were constructed using the matrices $(\boldsymbol{M}_j)_{j \neq i}$ only. The predictions so obtained are referred to as *out-of-fold predictions* in machine learning parlance.

Model fit was evaluated using the root mean squared error (RMSE) between the true outcome values and the out-of-fold predictions.

By way of a baseline, I used training set mean imputation. Precisely, I computed

$$\widehat{y}_i = \frac{1}{19} \sum_{j \neq i} y_j,$$

for $i = 1, \ldots, 20$.

Distance computation and kernel construction. Several distances between the input matrices were computed.

First, I computed $\widehat{\mathbf{SW}}_2^2(\boldsymbol{M}_i, \boldsymbol{M}_j; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_{200})$ for all $1 \leq i < j \leq 20$ in order to build the corresponding kernel $\hat{k}_{SW}(\boldsymbol{M}_i, \boldsymbol{M}_j; \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_{200})$. The same realisations of $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_{200} \overset{\text{i.i.d.}}{\sim} \sigma$ were reused for all sliced Wasserstein computations. I also computed Wasserstein distances along the cardinal axes in order to build the kernel $k(\boldsymbol{M}_i, \boldsymbol{M}_j; \boldsymbol{e}_1, \ldots, \boldsymbol{e}_4), 1 \leq i < j \leq 20$ (see Definition 3.12 on page 23).

Next, I computed sliced Wasserstein distances between scaled versions of the input matrices. More precisely, when predicting the i-th outcome, I computed a pooled estimate of the joint covariance matrix of $(\widehat{P}_j)_{j \neq i}$ as

$$\widehat{\boldsymbol{\Sigma}}_i = \frac{1}{19} \sum_{j \neq i} \widehat{\boldsymbol{\Sigma}}(\boldsymbol{M}_j),$$

where $\widehat{\mathbf{\Sigma}}(P_j)$ is the usual estimate of the covariance matrix of the j-th distribution based on the empirical sample represented in $\boldsymbol{M}_j$. The input matrices were then scaled as

$$\widetilde{\boldsymbol{M}}_k = \boldsymbol{M}_k \boldsymbol{S}_i,$$

with $\boldsymbol{S}_i$ the diagonal matrix containing the multiplicative inverses of the square roots of the diagonal elements of $\widehat{\mathbf{\Sigma}}_i$, $k = 1, \ldots, 20$. So while the i-th input matrix $\boldsymbol{M}_i$ was scaled, the scaling matrix was not learnt from it. Note that no centring was applied as the theoretical considerations from this chapter showed them to be irrelevant for sliced Wasserstein computations. I computed the Wasserstein distances between the scaled input matrices as $\widehat{\mathbf{SW}}_2^2(\boldsymbol{M}_i, \boldsymbol{M}_j; \boldsymbol{S}_k^\top \boldsymbol{\theta}_1, \ldots, \boldsymbol{S}_k^\top \boldsymbol{\theta}_{200})$ in order to build the corresponding kernel for predicting $y_k, k = 1, \ldots, 20$. The Wasserstein distances along the cardinal axes were not recomputed for the scaled input matrices since global scaling merely amounts to picking a different length-scale for the Wasserstein kernel.

Finally, I applied global whitening transformations to the input matrices. Using the spectral decomposition (and dropping the index for ease of exposition)

$$\widehat{\mathbf{\Sigma}} = \boldsymbol{U}\mathbf{\Lambda}\boldsymbol{U}^\top,$$

two whitening matrices were defined as

$$\begin{aligned} \boldsymbol{W}_{\text{ZCA}} &= \boldsymbol{U}\mathbf{\Lambda}^{-1/2}\boldsymbol{U}^\top, \\ \boldsymbol{W}_{\text{PCA}} &= \mathbf{\Lambda}^{-1/2}\boldsymbol{U}^\top. \end{aligned}$$

The whitening matrices $\boldsymbol{W}_{\text{ZCA-cor}}$ and $\boldsymbol{W}_{\text{PCA-cor}}$ were defined similarly, but starting from the estimated correlation matrix rather than the estimated covariance matrix. More precisely, we use the spectral decomposition of the correlation matrix

$$\boldsymbol{S}^{-1}\widehat{\mathbf{\Sigma}}\boldsymbol{S}^{-1} = \boldsymbol{V}\Delta\boldsymbol{V}^\top$$

and define

$$\begin{aligned} \boldsymbol{W}_{\text{ZCA-cor}} &= \boldsymbol{V}\mathbf{\Delta}^{-1/2}\boldsymbol{V}^\top\boldsymbol{S}^{-1}, \\ \boldsymbol{W}_{\text{PCA-cor}} &= \mathbf{\Delta}^{-1/2}\boldsymbol{V}^\top\boldsymbol{S}^{-1}. \end{aligned}$$

A fifth whitening matrix was defined as

$$\boldsymbol{W}_{\text{Chol}} = \boldsymbol{L}^{-1},$$

where $\boldsymbol{L}$ is the lower triangular matrix resulting from the Cholesky decomposition $\widehat{\mathbf{\Sigma}} = \boldsymbol{L}\boldsymbol{L}^\top$. Starting from $\widehat{\mathbf{\Sigma}}$, the whitening matrices were constructed using the R package `whitening` (Strimmer et al., 2022).

For the sliced Wasserstein distances, theory shows that the choice of the precise whitening procedure is not important: While, for fixed $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_{200}$, $\widehat{\mathbf{SW}}_2^2(\boldsymbol{M}_i, \boldsymbol{M}_j; \boldsymbol{W}^\top\boldsymbol{\theta}_1, \ldots, \boldsymbol{W}^\top\boldsymbol{\theta}_{200})$ is *not* invariant to the choice of a whitening matrix derived from $\widehat{\mathbf{\Sigma}}$, differences in the result are due solely to sampling error. I therefore arbitrarily picked $\boldsymbol{W}_{\text{ZCA}}$ as the whitening matrix and only estimated the sliced Wasserstein distances as $\widehat{\mathbf{SW}}_2^2(\boldsymbol{M}_i, \boldsymbol{M}_j; \boldsymbol{W}_{\text{ZCA}}^\top \boldsymbol{\theta}_1, \ldots, \boldsymbol{W}_{\text{ZCA}}^\top \boldsymbol{\theta}_{200})$. For the Wasserstein kernels, however, all five methods were applied. That is, I constructed

$$k(\boldsymbol{M}_i, \boldsymbol{M}_j; \boldsymbol{W}^\top \boldsymbol{e}_1, \ldots, \boldsymbol{W}^\top \boldsymbol{e}_4),$$

$1 \leq i < j \leq 20$, for each $\boldsymbol{W} \in \{\boldsymbol{W}_{\text{ZCA}}, \boldsymbol{W}_{\text{PCA}}, \boldsymbol{W}_{\text{ZCA-cor}}, \boldsymbol{W}_{\text{PCA-cor}}, \boldsymbol{W}_{\text{Chol}}\}$.

The kernels so constructed were fed to separate Gaussian process regression models, one for each kernel, for each outcome variable, and for each of the twenty folds. The hyperparameters were tuned as explained in Chapter 3, separately for each kernel, for each outcome variable, and for each of the twenty folds. Hyperparameter searches were initialised randomly fifty times.

5.6.3 Results

Figure 5.5 on the next page plots the out-of-fold predictions for the twenty 10% trimmed means for each modelling approach. A baseline model that does not take into account the input matrices achieves an RMSE of 0.759. The model based on the sliced Wasserstein distances between the raw input matrices does not outperform this baseline; the model based on the sliced Wasserstein distances between the scaled input matrices fares slightly better with an RMSE of 0.697. Whitening improves performance more noticeably for the sliced Wasserstein kernel, resulting in an RMSE of 0.601.

Kernels based on the Wasserstein distances along the cardinal axes, labelled 'marginal Wasserstein' in the figure, tend to perform better in this example. When based on the raw input matrices, performance is modest at an RMSE of 0.652. However, whitening using the ZCA-cor, ZCA or Cholesky approach roughly half this RMSE. Wasserstein kernels based on PCA- and PCA-cor-whitened input matrices show in-between performance.

Figure 5.6 on page 54 shows the out-of-fold predictions for entropy outcomes. A baseline model that does not take into account the input matrices achieves an RMSE of 0.165, which neither the sliced nor the 'marginal' Wasserstein kernels based on raw input matrices are able to beat. Scaling slightly improves the performance of the sliced Wasserstein kernel (RMSE: 0.153); whitening results in a further improvement still (RMSE: 0.126).

The effects of whitening are more pronounced for the 'marginal' Wasserstein kernels, with ZCA-cor-whitening resulting in RMSEs of 0.063 but the other whitening approaches trailing noticeably.

5.6.4 Conclusions

First, the results obtained on this more realistic dataset illustrate the effect that the global affine operations of scaling and whitening have on the usefulness of sliced Wasserstein distances for regression tasks: for both outcome variables, predictive performance was quite literally worse than useless when using sliced Wasserstein distances between the raw input matrices. After scaling, the sliced Wasserstein approach was able to beat the baseline; after whitening, the improvement was more noticeable still.

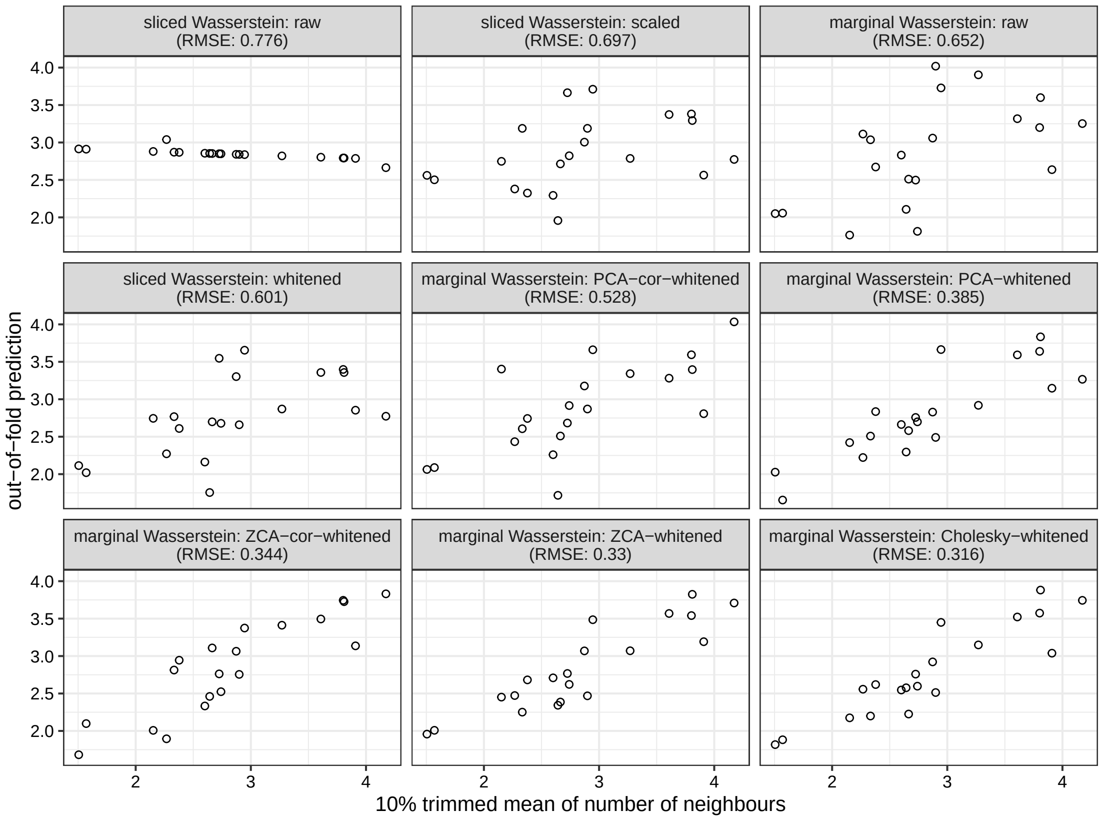

Figure 5.5: Out-of-fold predictions for 10% trimmed means of the number of neighbours by nine models based on different kernels. The facets are sorted left to right, top to bottom, in order of descending root mean squared error. The root mean squared error of a baseline model that uses the mean of the in-fold outcomes as the prediction was 0.759, meaning that sliced Wasserstein kernel based on the raw input matrices could not beat this baseline.

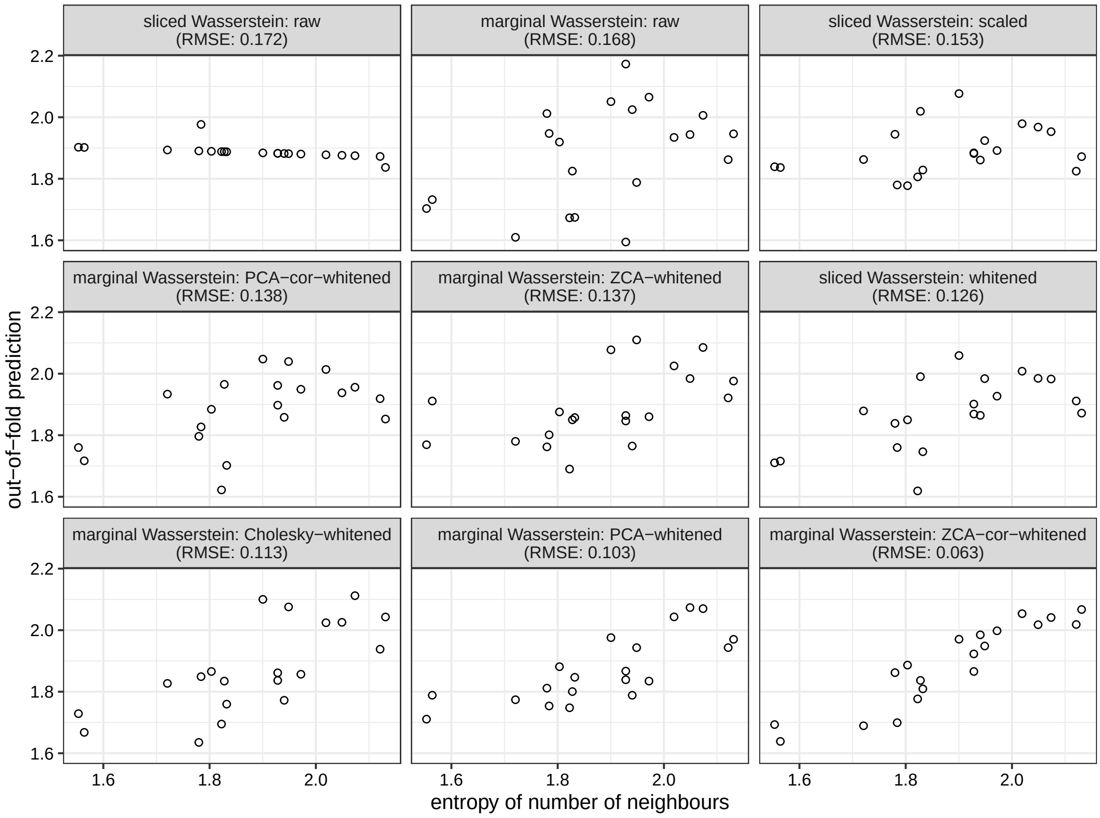

Figure 5.6: Out-of-fold predictions for the entropy of the empirical distributions of the number of neighbours by nine models based on different kernels. The facets are sorted left to right, top to bottom, in order of descending root mean squared error. The root mean squared error of a baseline model that uses the mean of the in-fold outcomes as the prediction was 0.165, meaning that both the sliced and marginal Wasserstein kernels based on the raw input matrices could not beat this baseline.

Second, the results illustrate that the choice of the precise whitening method (shown theoretically not to affect sliced Wasserstein distances) may matter quite substantially when using Wasserstein distances along the cardinal axes. Unlike when using sliced Wasserstein distances, the orientation of the whitened coordinate system affects the Wasserstein distances along its cardinal axes.

Third, like in the more artificial example in the previous section, the results of the current section demonstrate that a modelling approach based on the sum of kernels derived from Wasserstein distances along the cardinal axes can compare favourably to a sliced Wasserstein approach. The insights from this and the preceding chapters suggest that this is because the relevant signal lives approximately along one or more of the cardinal axes in the (whitened) coordinate system.

5.7 Conclusions

The central theoretical insight presented in this chapter is that global affine transformations of the input distributions amount to a distortion of the measure on the unit sphere in the integral that defines the sliced Wasserstein distance. This distortion is characterised by a Radon–Nikodym derivative that depends only on $\boldsymbol{A}^\top \boldsymbol{A}$, and not on the transformation matrix $\boldsymbol{A}$ itself or on any global translation. In practical terms, this means that analysts wishing to apply a global whitening transformation do not need to consider the pros and cons of different whitening strategies: up to sampling error, they all give rise to the same sliced Wasserstein distances. This, importantly, is not the case for Wasserstein distances computed along individual directions; the real-data example illustrates that the choice of whitening procedure can substantially affect the usefulness of such distances.

An asymptotic analysis further showed that extremely anisotropic transformations cause uninformative or degenerate kernels. Hyperparameter tuning can be used to combat these effects. A practical recommendation is to always tune hyperparameters when applying anisotropic transformations so as to avoid degraded performance.

The simulations also revealed that, when the signal is axis-aligned, kernels based on Wasserstein distances along the cardinal directions can match or outperform sliced Wasserstein kernels. The real-data example corroborated this finding, which motivates the use of such kernels as a benchmark in future work.

Chapter 6

Conclusions and outlook

I set out to answer these two questions (see page 2):

1. Can examples be identified in which the combination of sliced Wasserstein distances and Gaussian process regression substantially outperforms natural alternative approaches even outside of the graph domain covered by Carpintero Perez et al. (2024, 2025)?

2. Rescaling a single dimension can drastically alter pairwise sliced Wasserstein distances, making them scale-sensitive in a way that many vector-based methods are not. How should analysts preprocess distributions to address this sensitivity, and can predictive accuracy be improved by deliberately rescaling informative dimensions?

As Chapter 4 has demonstrated, the answer to the first question is 'yes': If the marginal distributions are ambiguous as to the outcome variable but the joint distribution is not, the combination of sliced Wasserstein distances and Gaussian process regression handily outperforms natural alternatives based on the marginal distributions. That said, the examples in Chapter 5 illustrate that, in other situations, this same combination can struggle to perform on par with approaches based only on marginal distributions.

This observation suggests that it may be useful to consider both differences between the input distributions' marginals as well as differences between their copulas. One concrete way to achieve this would be to build kernels based on Wasserstein distances between the marginal distributions and to supplement these with a kernel that quantifies differences between the input distributions' copulas, for instance by converting the marginals to uniform distributions before computing sliced Wasserstein distances. Such an approach may enable analysts to pinpoint more accurately which aspects of the input distributions carry the predictive signal, which could allow them to expend modelling and computational effort in a more targeted way: After all, if the signal is contained almost entirely in a handful of margins, it seems wasteful to evaluate distances to new inputs along redundant directions. Of course, the statistical and mathematical implications of this proposal would have to be studied in earnest, too.

As for the second question, Chapter 5 showed that the effect of global affine transformations of the input distributions on sliced Wasserstein distances amounts to a distortion of the measure on the unit sphere. For an affine transformation $\boldsymbol{x} \mapsto \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}$, with $\boldsymbol{x}, \boldsymbol{b} \in \mathbb{R}^d$, $\boldsymbol{A} \in \mathrm{GL}_d(\mathbb{R})$, this distortion depends solely on $\boldsymbol{A}^\top \boldsymbol{A}$, not on $\boldsymbol{A}$ itself. A practical consequence of this is that, when whitening the input distributions, the precise choice of whitening procedure does not affect the resulting sliced Wasserstein distances. The asymptotic analyses in this chapter also showed that extremely anisotropic affine transformations of the input distributions lead to degenerate kernel matrices, even if they emphasise the linear subspaces that carry the predictive signal. Appropriate hyperparameter tuning offsets this effect.

To wrap up this thesis, I outline some avenues for future research that may make for interesting Master's thesis topics.

The first concerns the role of the kernel function to which the (sliced) Wasserstein distances are fed. I only considered kernels based on the Gaussian radial basis function; an analysis of the usefulness of other kernel functions, such as the power exponential and Matérn covariance functions, would be a natural next step. Moreover, it remains to be seen if there are circumstances in which the computationally more demanding kernel $\hat{k}_{USW}$ (see page 23) is more useful for predictive modelling than is $\hat{k}_{SW}$.

We have seen that a global affine transformation of the input distributions by means of a matrix $\boldsymbol{A} \in \mathrm{GL}_d(\mathbb{R})$ corresponds to a distortion of the measure in the integral in the definition of the sliced Wasserstein distance. The second avenue for future work concerns the questions of whether and how a suitable matrix $\boldsymbol{A}$ can be learnt efficiently from the training data. Relatedly, I did not have any success in using importance sampling to obtain an estimate of $\mathbf{SW}_2^2(\boldsymbol{M}_i\boldsymbol{A}, \boldsymbol{M}_j\boldsymbol{A})$ from $\left(\mathbf{W}_2^2(\langle \boldsymbol{M}_i, \boldsymbol{\theta}_\ell \rangle, \langle \boldsymbol{M}_j, \boldsymbol{\theta}_\ell \rangle)\right)_{\ell=1}^{L}$ (Remark 5.7 on page 41). But a more targeted analysis may yet yield some useful insights.

Finally, throughout this thesis, the focus has been exclusively on solving regression problems by means of Gaussian process models. An exploration of how Wasserstein and sliced Wasserstein distances can be used to solve classification problems and how they can be incorporated in other regression models would naturally extend the results developed in this thesis.

References

Bachoc, François, Alexandra Suvorikova, David Ginsbourger, Jean-Michel Loubes & Vladimir Spokoiny. 2020. Gaussian process with multidimensional distribution inputs via optimal transport and Hilbertian embedding. *Electronic Journal of Statistics* 14. 2742–2772. doi:10.1214/20-EJS1725.

Bonnotte, Nicolas. 2013. *Unidimensional and evolution methods for optimal transportation*: Université Paris Sud - Paris XI; Scuola normale superiore (Pisa) dissertation. `https://theses.hal.science/tel-00946781v1`.

Breiman, Leo. 2001. Random forests. *Machine Learning* 45(1). 5–32. doi:10.1023/A:1010933404324.

Carpintero Perez, Raphaël, Sébastien Da Veiga, Josselin Garnier & Brian Staber. 2024. Gaussian process regression with sliced Wasserstein-Weisfeiler-Lehman graph kernels. *Proceedings of Machine Learning Research* 238. 1297–1305.

Carpintero Perez, Raphaël, Sébastien Da Veiga, Josselin Garnier & Brian Staber. 2025. Learning signals defined on graphs with optimal transport and Gaussian process regression. *Proceedings of Machine Learning Research* 258.

Dowson, D. C. & B. V. Landau. 1982. The Fréchet distance between multivariate normal distributions. *Journal of Multivariate Analysis* 12. 450–455. doi:10.1016/0047-259X(82)90077-X.

Eckermann, Marina, Bernhard Schmitzer, Franziska van der Meer, Jonas Franz, Ove Hansen, Christine Stadelmann & Tim Salditt. 2021. Three-dimensional virtual histology of the human hippocampus based on phase-contrast computed tomography. *PNAS* 118(48). e2113835118. doi:10.1073/pnas.2113835118.

Kessy, Agnan, Alex Lewin & Korbinian Strimmer. 2018. Optimal whitening and decorrelation. *The American Statistician* 72(4). 309–314. doi:10.1080/00031305.2016.1277159.

Klein, Nadja. 2024. Distributional regression for data analysis. *Annual Review of Statistics and Its Application* 11. 321–346. doi:10.1146/annurev-statistics-040722-053607.

Marples, Callum Robert & Philip Michael Williams. 2024. Patch area and uniform sampling on the surface of any ellipsoid. *Numerical Algorithms* 95. 1801–1827. doi:10.1007/s11075-023-01628-4.

Meunier, Dimitri, Massimiliano Pontil & Carlo Ciliberto. 2022. Distribution regression with sliced Wasserstein kernels. *Proceedings of Machine Learning Research* 162. 15501–15523.

Nadjahi, Kimia. 2021. *Sliced-Wasserstein distance for large-scale machine learning: Theory, methodology and extensions*: Institut Polytechnique de Paris dissertation. `https://theses.hal.science/tel-03533097`.

Nguyen, Khai. 2025. *An introduction to sliced optimal transport: Foundations, advances, extensions, and applications*. doi:10.48550/arXiv.2508.12519.

Nocedal, Jorge & Stephen J. Wright. 2006. *Numerical optimization*. Springer 2nd edn. doi:doi.org/10.1007/978-0-387-40065-5.

Peyré, Gabriel & Marco Cuturi. 2019. Computational optimal transport with applications to data sciences. *Foundations and Trends in Machine Learning* 11. 355–607. doi:10.1561/2200000073.

Poczos, Barnabas, Aarti Singh, Alessandro Rinaldo & Larry Wasserman. 2013. Distribution-free distribution regression. *Proceedings of Machine Learning Research* 31. 507–515.

R Core Team. 2025. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing Vienna, Austria. `https://www.R-project.org/`.

Rabin, Julien, Gabriel Peyré, Julie Delon & Marc Bernot. 2012. Wasserstein barycenter and its application to texture mixing. In Alfred M. Bruckstein, Bart M. ter Haar Romeny, Alexander M. Bronstein & Michael M. Bronstein (eds.), *Scale space and variational methods in computer vision*, 435–446. doi:10.1007/978-3-642-24785-9_37.

Rachev, Svetlozar T. & Ludger Rüschendorf. 1998. *Mass transportation problems, volume 1: Theory*. Springer. doi:10.1007/b98893.

Rasmussen, Carl Edward & Christopher K. I. Williams. 2006. *Gaussian processes for machine learning*. MIT Press. doi:10.7551/mitpress/3206.001.0001.

Riesen, Kaspar. 2025. Graph-based pattern recognition. Lecture notes, University of Bern. Version 3.0.

Schoenberg, Isaac Jacob. 1938. Metric spaces and positive definite functions. *Transactions of the American Mathematical Society* 44(3). 522–536. doi:10.1090/S0002-9947-1938-1501980-0.

Strimmer, Korbinian, Takoua Jendoubi, Agnan Kessy & Alex Lewin. 2022. *whitening: Whitening and high-dimensional canonical correlation analysis*. doi:10.32614/CRAN.package.whitening. `https://CRAN.R-project.org/package=whitening`. R package version 1.4.0.

Villani, Cédric. 2009. *Optimal transport: Old and new*. Springer. doi:10.1007/978-3-540-71050-9.

Declaration of consent

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